

CHAPTER ONE

Introduction

1.1 Background

Large Language Models (LLMs) have proven themselves capable of performing an impressive range of natural language processing tasks, such as text generation, question answering, summarization, and conversation. Models like GPT-4, LLaMA, and PaLM have reached state-of-the-art levels in many benchmark tests due to extensive pre-training on large bodies of text (Brown et al., 2020). At the same time, LLMs are limited by their parameters in terms of knowledge. In other words, when asked questions concerning data not included in the training dataset, outdated facts, or specific factual knowledge, LLMs often produce plausible but factually incorrect answers, a “behaviour” known as hallucinations (Ji et al., 2023).

Hallucination constitutes a major barrier to the adoption of LLMs in knowledge-heavy domains such as healthcare, legal research, scientific inquiry, and enterprise knowledge management systems, where factual correctness is crucial and not optional. The inadequacy of parametric models in differentiating between what they know and what they do not know has led to the creation of systems that augment generation with external knowledge retrieval.

The concept of Retrieval-Augmented Generation (RAG), created by Lewis et al. (2020), is a model that combines neural document retrieval and sequence-to-sequence generation. In a RAG pipeline, when a user submits a query, the retrieval component searches an external document corpus to identify text chunks that are likely to contain relevant information. These retrieved text chunks are then brought forward as additional context to the generative model, which bases its output on both the original query and the retrieved evidence. This design reduces the model’s dependence on just its parametric memory and grounds generation in verifiable external sources.

The original RAG architecture showed that relatively small retrieval-augmented models can perform better than much larger standalone language models on tasks involving factual knowledge such as open domain question answering, fact verification, and knowledge-grounded dialogue (Lewis et al., 2020). There have been many advancements made since then to expand upon the RAG approach, some of which include better retriever pretraining methods (Karpukhin et al., 2020), architectures supporting multi-document fusion like Fusion-in-Decoder (Izacard &

Grave, 2021), retrieval-augmented pretraining using REALM (Guu et al., 2020), and scaling retrieval to trillion-token corpora with RETRO (Borgeaud et al., 2022). These developments have established RAG as a foundational paradigm for building knowledge-grounded, factually reliable language systems.

1.2 The Role of Retrieval in RAG Systems

The effectiveness of a RAG system is largely dependent on the quality of its retrieval component. If the retriever fails to retrieve relevant documents, there will be no factual knowledge for the generator to produce correct answers, regardless of its own language modelling capabilities. The relationship described above means that retrieval quality plays a key role in determining the overall effectiveness of the RAG system.

A standard RAG pipeline involves several retrieval design decisions that can significantly affect downstream output quality. These may include the choice of retrieval method (sparse, dense, or hybrid), the application of the reranking model in order to improve the initial retrieval; the number of retrieved documents sent to the generator (top-k selection); and the fusion of different retrieval results. These decisions are retrieval configuration choices, and the interaction between these choices determines the quality of context available to the generator to give the output (response).

Dense retrievers, including Dense Passage Retrieval (DPR) (Karpukhin et al., 2020), encode the queries and document passages into a dense vector space and retrieve document passages based on semantic similarity. The advantage of dense retrieval is that it allows capturing semantic relationships between queries and passages regardless of their lexical overlap. However, dense retrievers can struggle with rare entities, exact keyword matches, and out-of-distribution queries.

Sparse retrieval methods such as BM25 (Robertson & Zaragoza, 2009) rely on lexical matching and frequency of terms. Despite being inferior to dense retrieval in terms of semantic flexibility, sparse retrievers are very efficient when it comes to queries that require an exact term match, and remain competitive baselines in many retrieval benchmarks.

Hybrid retrieval combines dense and sparse retrieval to take full advantage of the complementary strengths of both approaches. A common fusion technique is Reciprocal Rank Fusion (RRF), that merges ranked lists of multiple retrievers using their reciprocal ranks,

producing a unified ranking that usually performs better than either individual method (Cormack et al., 2009). Modern RAG implementations widely use hybrid retrieval, but its particular contribution to factual accuracy has not been studied enough.

Re-ranking introduces a second-stage model, usually a cross-encoder, that co-encodes the query and each candidate document to obtain a more precise score for their relevancy. Unlike bi-encoders, cross-encoders are capable of capturing interaction between tokens from queries and documents, hence being able to detect slight differences in the level of relevance. Re-rankers are proven to significantly increase the precision of the retrieved information (Nogueira & Cho, 2019), and their usage in RAG is becoming more widespread. Nevertheless, many studies of the re-ranking effect assess it based on retrieval metrics like Mean Reciprocal Rank (MRR) or Recall@K instead of assessing the improvement in the accuracy of generated answers.

1.3 Research Gap

In recent times, there has been a great deal of research that focuses on the retrieval quality in RAG models. Recent papers discuss new architectures of the re-ranker (Dong et al., 2024), create a platform for benchmarking re-rankers through human and LLM-based evaluations (Abdallah et al., 2025), propose the utilization of utility-based retrieval assessment methods that better predict the RAG performance (Salemi & Zamani, 2024), as well as provide a taxonomy of RAG evaluation dimensions (Yu et al., 2024).

Nonetheless, a notable finding from the existing literature is the fact that research usually focuses on the improvement of retrieval or re-ranking components alone, or comparison of re-rankers against each other. The relatively less explored research aspect is the comparison of retrieval configuration methods through controlled evaluation of how dense retrieval, hybrid fusion, and re-ranking influence the factual correctness of the generated answers in the same RAG pipeline.

For instance, the G-RAG framework proposed by Dong et al. (2024) is a graph-based re-ranker that uses connections between documents for enhancing the ranking ability. In their evaluations, they show improvements in ranking metrics compared to baseline re-rankers, but they do not evaluate the effect of the changes on the generation-level factual correctness. Also, Abdallah et al. (2025) came up with RankArena benchmarking platform that assesses 24 re-ranking methods across 84 models using pairwise preference and LLM-as-a-judge evaluation. While RankArena

provides valuable tools for re-ranker comparison, its focus is on ranking performance rather than on how re-ranking configurations affect the factual reliability of generated outputs.

It is worth mentioning that Salemi & Zamani (2024) introduce a very useful methodological contribution through their work with eRAG, evaluating the quality of retrieval through the measurement of the contribution of each separate document on the downstream performance of LLMs. Their results prove the poor predictability of the traditional relevance labels on the factual quality of RAG outputs and support the thesis that the utility-based evaluation is much more effective at capturing the real value of the retrieval process. However, the authors do not compare the particular configurations of retrievals as independent variables.

However, this problem is important since practitioners who build RAGs have to make the decision whether to implement dense retrieval, or additionally use sparse retrieval with the fusion technique, or add re-ranking, or all of these at once, while there is no empirical evidence of the effect of these configurations on factual accuracy. The absence of controlled comparative studies means these decisions are often made on the basis of retrieval-level metrics alone, without direct evidence of their effect on generation correctness.

1.4 Problem Statement

It is still not clear enough through empirical studies how much each of the two techniques of retrieval fusion and re-ranking influence factual accuracy of Retrieval-Augmented Generation systems. The majority of existing studies on the topic either examine retrieval and re-ranking modules separately or compare re-rankers themselves, but do not conduct systematic comparison of how retrieval settings affect the factual accuracy of the output through the pipeline. There exists a difference between retrieval evaluation and generation-level evaluation, which this study intends to bridge.

1.5 Research Aim and Objectives

1.5.1 Aim

The aim of this research is to quantitatively evaluate the influence of retrieval enhancement techniques on factual accuracy of Retrieval-Augmented Generation systems.

1.5.2 Objectives

To achieve this aim, the study pursues the following objectives:

Objective 1: Develop a baseline dense retrieval RAG system using an established open-domain question answering benchmark.

Objective 2: Implement hybrid fusion retrieval and cross-encoder re-ranking within the same controlled RAG pipeline.

Objective 3: Compare four retrieval configurations; dense-only, dense with re-ranking, hybrid fusion, and hybrid fusion with re-ranking, for their impact on retrieval quality, factual accuracy, and faithfulness.

1.6 Research Questions

The study is guided by the following research questions:

RQ1: How does hybrid fusion retrieval affect factual accuracy compared to dense retrieval alone in RAG systems?

RQ2: To what extent does re-ranking improve downstream factual accuracy under fixed retrieval conditions?

RQ3: Does combining fusion and re-ranking produce greater gains in factual accuracy than applying either strategy independently?

1.7 Significance of the Study

This research fills the gap in the growing literature on the evaluation of retrieval-augmented generation (RAG) systems by investigating the relationship between benchmarks for retrieval and generation levels of analysis. Even though existing literature clearly highlights the role of the quality of retrieval in the effectiveness of RAG systems, not many empirical studies exist comparing the impact of different configurations of retrieval via generation-level evaluation.

In terms of practice, the present research provides evidence that can be useful in decision-making with regard to designing RAGs. Currently, the choice of configurations such as dense retrieval, hybrid fusion, and re-ranking largely depends on retrieval ranking metrics that were found to have a poor correlation with downstream generation quality by Salemi and Zamani (2024).

In this research, a dual level evaluation approach is adopted, where both retrieval and generation

quality are evaluated. Retrieval quality is evaluated using Recall@5, MRR@5, and NDCG@5 scores. Meanwhile, the generation quality is evaluated along two dimensions – factual accuracy, using answer correctness with respect to gold standard answers in terms of exact match and token-level F1 score, and faithfulness using the RAGAS framework which evaluates the groundedness of the answers generated from the retrieved passages. The importance of this distinction comes from the fact that answer correctness and faithfulness are two different constructs: an answer can be factually wrong but grounded, and conversely it can be factually right but ungrounded. Evaluating both in addition to the retrieval metrics allows this study to test whether the improvement in ranking translates to improvement in the factual quality of generated answers.

The empirical findings of this study offer guidance for designing better retrieval pipelines for real-world RAG applications.

1.8 Scope and Limitations

This study deals specifically with the retrieval configuration as the independent variable, analysed across dense only, hybrid fusion, re-ranking, and combinations thereof. The generative model remains the same across all conditions to focus on the effects of the retrieval configuration. This work does not propose novel retrieval or re-ranking algorithms; rather, existing publicly available components were tested in combinations.

For the performance evaluation, we used Natural Questions (Kwiatkowski et al., 2019), an open-domain question answering dataset adapted into BEIR framework (Thakur et al., 2021). The Natural Questions dataset is a popular open-domain question answering dataset with human relevance judgments for retrieval evaluation, gold standard short answers for the answer correctness evaluation, and the possibility of generative evaluation. This dataset is relevant for the stated research questions, but the results obtained from the use of this dataset might not apply in other domains or to other tasks. Extending the experiments to other datasets (TriviaQA, HotpotQA, FEVER) is suggested as future work.

This study uses only one re-ranker model (pre-trained cross-encoder), instead of evaluating different re-rankers models architectures. This limitation was made intentionally because the research question asks about the effect of re-ranking as a configuration on factual consistency,

while the comparison of performance between re-ranker models was already discussed in detail (Abdallah et al., 2025; Dong et al., 2024).

1.9 Structure of the Dissertation

The rest of this dissertation is structured as follows:

Chapter 2: Literature Review offers a comprehensive review of related literature, discussing the fundamentals of RAG, dense and hybrid retrieval models, re-ranking techniques, fusion methods, and RAG system evaluations.

Chapter 3: Methodology outlines the methodology of the study, detailing experimental settings, evaluation measures, data selection, and implementation aspects.

Chapter 4: Results Discussion presents the experimental findings across all four retrieval configurations, reporting both retrieval-level and generation-level metrics and interprets them in relation to the research questions and existing literature.

Chapter 5: Summary, Conclusions and Recommendations summarises the key findings, contributions, and directions for future work.

CHAPTER TWO

Literature Review

2.1 Foundations of Retrieval-Augmented Generation

The emergence of Retrieval-Augmented Generation resulted from an intrinsic weakness of large language models. Models like GPT-3 showed that sufficiently trained language models can be capable of demonstrating outstanding performance on a variety of natural language tasks without any task-specific fine-tuning and thus established the paradigm of few-shot learning via parametric knowledge alone (Brown et al., 2020). However, this dependency on the knowledge encoded in the parameters of the model during pre-training has led to one important issue: when asked questions about the information not present in the training set, about events that happened later than the moment of training completion, or about anything else that would require factual recall, large language models often produce plausible but factually wrong answers. This problem known as hallucination is a major obstacle on the way to using language models in knowledge-intensive applications (Ji et al., 2023).

RAG, or Retrieval-Augmented Generation, was proposed by Lewis et al. (2020) as an architectural solution to this problem. Instead of using only parametric memory, the RAG framework incorporates a retrieval module that allows querying an external document database during inference, thus providing the generative model with additional context in the form of relevant passages. In the original RAG setup, a passage retrieval step relied on a dense retriever, which used a bi-encoder trained on Wikipedia documents, and a sequence-to-sequence generator conditioned its outputs on the input query and the retrieved passages. Lewis et al. showed that this architecture allowed obtaining superior performance for retrieval-augmented models on question answering and fact verification benchmarks compared to much larger standalone language models, establishing retrieval augmentation as a viable alternative to continued scaling of parametric knowledge.

Further research expanded on the RAG framework in multiple ways. Guu et al. (2020) developed REALM, which added the step of retrieval directly to the pre-training phase of the framework, instead of using it as an additional step in inference time. By pre-training both retriever and generator, REALM enabled the model to be capable of performing retrieval in such a way that

would optimize its performance on downstream tasks, showing how retrieval-augmented pre-training could increase factual accuracy of question answering models on open-domain question answering tasks. Izacard and Grave (2021) presented Fusion-in-Decoder (FiD), which extended the capabilities of RAG by enabling fusion of multiple documents by encoding each of them separately and then concatenating their encodings in the decoder of the model, thereby allowing the generator to utilize information from many more retrieved documents simultaneously, rather than only concatenating them as plain text. Borgeaud et al. (2022) took the idea of retrieval-augmented language models to a whole new scale with RETRO, which was capable of retrieving from a memory containing up to two trillion tokens, reducing the perplexity of language models and improving their factual accuracy without proportionally increasing the number of parameters.

Combining all these findings into one conclusion, it should be said that RAG became a paradigm of constructing knowledge-based language models, and the quality of retrieval was found out to play an important role in the overall performance of the language model. On the other hand, the specifics of the influence of retrieval configuration on generation were not sufficiently investigated in existing research, leading to the current systematic investigation in this paper.

2.2 Retrieval Methods in RAG

The quality of retrieved passages is an important factor of RAG performance. When retrieval does not manage to find relevant documents, generation will lack the necessary knowledge to produce correct results irrespective of the language modeling abilities. There are three main retrieval techniques used in RAG: dense retrieval, sparse retrieval, and hybrid retrieval technique combining both.

2.2.1 Dense Retrieval

The dense retrieval approach maps the queries and the documents into the same continuous vector space and perform document retrieval based on computing the similarity between the query and document vectors. The most notable dense model used for open-domain question answering tasks is called Dense Passage Retrieval (DPR), which was proposed by Karpukhin et al. (2020). The DPR model uses a dual encoder approach with two BERT-like encoders, where

each encodes either the query or the passage independently. The training process makes use of in-batch negatives and hard BM25 negative passages. Retrieval at inference time is performed using maximum inner product search via FAISS, enabling efficient search over large passage collections.

In Karpukhin et al. (2020), dense retrieval systems were found to greatly surpass lexical retrieval systems, once fine-tuned on question-passage pairs: DPR was ahead of BM25 by 9-19% in top-20 passage retrieval accuracy in four out of five studied datasets, with DPR achieving 65.2% top-5 accuracy on Natural Questions compared with 42.9% for BM25, it was shown that higher retrieval precision translates directly into better end-to-end QA accuracy, providing empirical evidence for the relationship between retrieval performance and generation performance. The paper further demonstrated that dense retrieval alone, without a lexical component, is sufficient for strong practical performance.

Dense retrieval is especially good in establishing semantic relations between queries and passages in cases where there is little similarity in lexical items used. It is therefore useful in natural language processing where the form of the query will be quite different from that of the related passage text. However, DPR and similar bi-encoder architectures encodes queries and passages separately in such a way that the relevance scores are determined through an inner product of the independently generated representations, without direct token-level interaction between queries and passages. It is therefore a limitation which requires that re-ranking is done as the second step, as described in Section 2.3.

2.2.2 Sparse Retrieval

Sparse retrieval methods score documents against queries using lexical matching, assigning weights to terms based on their frequency in a document and their discriminative value across the corpus. The leading sparse retrieval algorithm in both academic research and production information retrieval systems is BM25, which is based on the Probabilistic Relevance Model proposed by Robertson & Zaragoza (2009).

BM25 assigns scores to documents using a combination of a saturation function of term frequencies and inverse document frequencies. Soft document length normalisation is performed

to adjust for the different lengths of documents. BM25 is considered to be among the most successful text retrieval methods according to Robertson and Zaragoza (2009). The algorithm fits into the framework of the probabilistic model which attempts to calculate the probability of relevance for each query-document pair and rank documents accordingly. The term weighting scheme of BM25 is characterised by the usage of a saturation curve in order to take into consideration the diminishing returns of the repeated presence of a query term in a document, which Robertson & Zaragoza (2009) call 'saturation'. This distinguishes BM25 from simpler tf-idf variants where term frequency contributes linearly and without bound. Parameter guidance derived from accumulated retrieval experiments suggests values of $0.5 < b < 0.8$ for length normalisation and $1.2 < k_1 < 2$ for term frequency scaling perform well across a range of document collections (Robertson & Zaragoza, 2009).

BM25 and other sparse retrieval techniques work very well for query-passage pairs for which there is an importance of having an exact term overlap between the query and the passage, such as queries containing rare entities, technical terminology, or precise named references. Sparse retrieval techniques like BM25 require no training and return a deterministic and interpretable score that is computed through term statistics obtained directly from the corpus. Hence, sparse retrieval is a fast and reliable method that works well in many retrieval benchmarks because of its simplicity and robustness despite its inability to capture semantic relationships beyond lexical overlap.

The main problem with sparse retrieval lies in the fact that it is dependent on surface-form term match. A query written in a different way compared to a passage but still semantically similar may get a very low BM25 score despite the high relevance of the passage. The complementary failure mode of sparse retrieval compared to dense retrieval is a motivation for the hybrid techniques described in Section 2.2.3.

2.2.3 Hybrid Retrieval and Fusion

Hybrid retrieval is a combination of dense and sparse retrieval approaches to utilize their advantages and overcome failure modes in a combined ranking. Sparse retrieval focuses on precision and accuracy for retrieving relevant passages, and dense retrieval on semantic similarity for queries that are far from their passages in terms of common lexical elements. Using

hybrid retrieval will help find a set of candidates that neither approach can reliably identify individually.

The most used approach to rank aggregation is Reciprocal Rank Fusion (RRF) developed by Cormack et al. (2009). It is an unsupervised rank aggregation algorithm which merges several ranked lists using the sum of reciprocal rank positions for each document across all lists: $RRFscore(d) = \text{sum of } 1/(k + r(d))$, where k is a constant parameter equal to 60, determined via pilot experiments. Cormack et al. (2009) demonstrated that RRF "consistently yields better results than any individual system," outperforming Condorcet Fuse and CombMNZ across multiple TREC experiments and the LETOR 3 dataset. On LETOR 3, RRF achieved a MAP of 0.6051 against Condorcet's 0.5917, with all improvements over individual rankers significant at $p < 0.003$. One of the main advantages of RRF is that it does not need any special voting mechanism or any additional information; it works entirely based on ranks and not scores, overcoming the problem of score-scale incompatibility among different retrieval models.

This score incompatibility is a fundamental challenge for hybrid retrieval methods that attempt to combine raw scores rather than ranks. Dense retrieval systems use inner product similarity scores which are confined by vector normalization, while BM25 scores are unbounded term frequency statistics, thus having an incompatible score scale with dense vectors. Therefore, the combination of raw scores involves additional normalization steps and parameters which may show unpredictable results for different distributions of queries. RRF is immune to this problem, because it works exclusively with rank positions.

While there is evidence about the benefits of hybrid retrieval in the level of retrieval quality, there is a lack of research about the influence of hybrid retrieval on the generation quality of RAG models. Prior studies evaluate hybrid retrieval mainly in terms of retrieval metrics like recall and MAP. This gap motivates the controlled comparison of hybrid and dense-only configurations under generation-level evaluation in the present study.

2.3 Re-ranking in RAG

Re-ranking introduces a second retrieval stage in which an initial set of candidate passages, retrieved by a first-stage retriever such as DPR or BM25, is rescored using a more computationally intensive model capable of capturing finer-grained query-passage interactions.

The reason why re-ranking is needed is due to some architectural limitations of the bi-encoder dense retrievers, since DPR and similar models encode the queries and the passages individually, and the similarity score between the two is determined by the inner product of two separate embeddings, with no interaction at the token level. Re-ranking allows the use of cross-encoder models to obtain joint encoding of the query and candidate passage as one input sequence, which makes it possible to have attention over token-level interactions.

Nogueira and Cho (2019) established the foundational work behind cross-encoder reranking for passage retrieval showing that fine-tuning BERT as a binary relevance classifier for query-passage pairs significantly beats previous neural and lexical ranking techniques. Here, concatenation of the query and passage as a single input sequence of BERT followed by the output representation to generate the relevance score is performed. Nogueira and Cho (2019) found that BERT-based cross-encoder systems surpassed the previous state-of-the-art models by a wide margin for both considered tasks, obtaining a value of 36.5 MRR@10 on the MS MARCO development set using a BM25 baseline of 16.7, beating the previous state-of-the-art model by 27%. It is noteworthy that the above achievement was done using less than 0.3% of the data available in the MS MARCO training set.

Subsequent work has extended re-ranking in the context of RAG systems specifically. Dong et al. (2024) introduced the concept of G-RAG, which is a graph-based re-ranker that builds document graphs based on Abstract Meaning Representation and employs graph neural networks to establish connections between documents during re-ranking. In this way, the model can make use of relations between retrieved documents rather than evaluating each one separately. G-RAG surpasses existing re-ranking baselines on both Natural Questions and TriviaQA datasets. As noted by the authors themselves, "the existence of excessive ties among the ranking scores produced makes it difficult for pre-trained large language models to improve RAG via reranking." Nevertheless, it is also worth mentioning that like most of the literature on re-ranking, Dong et al. (2024) evaluate their model using ranking metrics only.

The problem of evaluation infrastructure was tackled by Abdallah et al. (2025), who introduced RankArena, an open-source platform with five evaluation modes: blind pairwise comparison,

manual full-list annotation, LLM-as-a-judge, BEIR benchmark evaluation, and end-to-end RAG output assessment. RankArena reported an average agreement between human and LLM of 74.2% across all the evaluated models, and found great variation in performance between different re-ranking methods, with the best of them achieving an average BEIR score of 52.8 against relatively low scores of the weak baselines. Abdallah et al. noted that traditional evaluation metrics based on static data and pre-defined metrics lack the ability to measure user-oriented aspects like the relevance of documents, utility of answers, or faithfulness in generation. Nevertheless, like G-RAG, RankArena is mostly concerned about ranking performance and not the effect of re-ranking on the factual groundedness of generated answers.

Thus, the existing literature on re-ranking shows that it significantly increases the accuracy of ranking passages compared to first-stage retrieval. However, one thing that needs investigation is the influence of better passage ranking on the factual accuracy of the generated answers in RAG.

2.4 Evaluation of RAG Systems

RAG systems pose unique challenges to evaluation that are different from both language model evaluation on its own and information retrieval evaluation on its own. A RAG system generates outputs that depend on two inter-dependent components, retrieval and generation, and each of these components may positively or negatively impact the quality of output individually. This means that a proper evaluation of RAG systems should involve metrics that measure both retrieval effectiveness and generation quality and their interplay. Recent literature has made a shift towards realisation that those two levels of evaluation cannot be treated as being equal and that improving retrieval-level metrics does not necessarily mean an improvement in generation-level quality.

2.4.1 Retrieval-Level Evaluation

Classical information retrieval evaluation metrics like Recall@K, Mean Reciprocal Rank, and Normalised Discounted Cumulative Gain measure the quality of passage sets that are retrieved with respect to human-annotated relevance judgments. These metrics have been commonly used for evaluation of the retrieval component of RAG systems. For example, the framework of BEIR

that is used for benchmarking different retrieval tasks on different domains uses fixed corpora, queries, and relevance judgments (Thakur et al., 2021).

Nevertheless, Salemi and Zamani (2024) have found that there is a major weakness in using traditional relevance scores in evaluation of RAG retrieval. In their paper, they proposed using each document's individual contribution to downstream task performance as its relevance label, rather than relying on human relevance annotations, since human annotations may mark a passage as relevant even when it does not contribute to a correctly generated answer, and vice versa. In order to overcome this problem, Salemi and Zamani (2024) introduced eRAG evaluation framework where each retrieved document is processed by the downstream LLM and the output produced is evaluated against ground truth labels, so that contribution of each individual document into downstream performance serves as its relevance score. The new method was found to be significantly correlated with downstream RAG performance compared to all baselines used, showing absolute improvement in Kendall's tau correlation between 0.168 and 0.494, depending on the dataset, and at the same time it is 2.468 times faster than full end-to-end evaluation and uses 50 times less GPU memory than end-to-end approach. Thus, this evidence supports the claim that retrieval evaluation in RAG setting should be done via downstream generation performance rather than retrieved passages ranking only.

2.4.2 Generation-Level Evaluation

Generation-level evaluation for RAG models evaluates the quality of text output by the generation model, based on the retrieved context, focusing particularly on facts accuracy, faithfulness to the retrieved context, and relevancy to the original query. The study by Yu et al. (2024) offers a taxonomy of RAG evaluation dimensions through the Auepora approach, where the generation evaluation can be categorized on three dimensions: relevancy of the generated answer to the original query, faithfulness of the generated answer to the retrieved context, and correctness of the generated answer in comparison with ground truth. Yu et al. (2024) observe that "most existing benchmarks address only one or a subset of these dimensions rather than the full evaluation surface," and explicitly identify the relationship between retrieval metrics and final generation outputs as an open research problem requiring further investigation.

Es et al. (2024) proposed RAGAS, which is a reference-free evaluation method that operationalizes generation-level RAG evaluation without necessitating any human-annotated ground truth answers. Three metrics are offered by RAGAS, namely Faithfulness, which measures whether claims in the generated answer are supported by the retrieved context, Answer Relevance, which evaluates whether the generated answer responds to the question, and Context Relevance, which evaluates whether the retrieved context is focused on and lacks any irrelevant information. In the WikiEval dataset, RAGAS achieved an agreement accuracy of 0.95 for Faithfulness, 0.78 for Answer Relevance, and 0.70 for Context Relevance compared to human annotators, far outperforming the GPT Score and GPT Ranking baselines in all three metrics. This framework has been developed in such a way that it is not dependent on the presence of human annotated answers to function, making it suitable for use in scenarios where such references are unavailable.

RAGAS Faithfulness is adopted in this study as a means of measuring the groundedness of answers generated based on retrieved context. Since faithfulness assesses if the answer generated is justified by the passages retrieved as opposed to checking for the factual correctness of the answer, it is coupled with measures of answer correctness (Exact Match and token-level F1) based on gold standard answers, as discussed in Chapter Three. This two-metric approach distinguishes groundedness from factual accuracy, which faithfulness alone does not capture.

2.4.3 The Gap Between Retrieval and Generation Evaluation

The one recurring observation made about RAG evaluation across studies is that retrieval level and generation level metrics do not vary hand-in-hand. Salemi and Zamani (2024) have shown through experimentation that relevance labels used traditionally are inaccurate predictors of RAG quality outputs; their findings show that while a passage can be high-scoring in terms of the retrieval metric, it does not necessarily result in an faithful or factual answer generation. According to Yu et al. (2024), there is still a gap to fill in understanding the association between the performance of retrieval techniques and generation results, and therefore, further research is necessary in that regard.

This is directly related to the current study, which attempts to bridge the gap between retrieval level and generation level evaluation in the evaluation process of retrieval configurations.

Previous research in the comparison of retrieval configurations, including that on re-ranking and hybrid fusion approaches, has almost entirely been based on evaluation using only retrieval level metrics, without accounting for the effect on answer generations.

2.5 Research Gap Synthesis

The literature discussed in this chapter provides well-established research in various aspects of RAG systems. The theoretical and empirical basis of retrieval-augmented generation is well-established, since the original RAG architecture (Lewis et al., 2020), retrieval-augmented pre-training (Guu et al., 2020), multi-document fusion (Izacard & Grave, 2021), and large scale retrieval (Borgeaud et al., 2022) prove that the generation of text based on external retrieval significantly improves factual accuracy compared to parametric methods only. Dense retrieval has already been established as a strong base for RAG systems (Karpukhin et al., 2020), hybrid fusion has been reported to improve retrieval recall over single-method retrieval in several settings (Cormack et al., 2009), and cross-encoder re-ranking has been proved to significantly improve passage retrieval precision than first stage retrieval (Nogueira & Cho, 2019). RAG evaluation methodologies have been developed up to distinguishing between retrieval quality and generation quality, as the recent eRAG framework (Salemi & Zamani, 2024) proves that retrieval metrics poorly correlate with downstream generation performance, and RAGAS (Es et al., 2024) provides reference-free generation-level evaluation framework.

There is, however, an evident and substantial gap that runs consistently through this literature review. Research on the role of retrieval configuration in RAGs has mostly been focused on assessing individual elements in isolation or comparing retrieval and re-ranking approaches using only retrieval metrics, without systematically measuring how different retrieval configuration choices affect the factual accuracy of generated answers within the same controlled pipeline. Dong et al. (2024) demonstrated improvements in ranking quality with graph-based re-ranking but explicitly did not evaluate end-to-end generation quality. Abdallah et al. (2025) proposed an infrastructure for testing the effectiveness of re-rankers using various ranking metrics but have failed to assess the impact of re-ranking configuration on downstream generation faithfulness. Salemi and Zamani (2024) have proven the need for downstream performance evaluation in case of retrieval but failed to compare different retrieval configurations as an independent variable. The connection between retrieval metrics and

generation output was highlighted by Yu et al. (2024) as an open question for future research, without resolving it empirically.

However, what the literature does not offer is a controlled and systematic comparison of dense retrieval, hybrid fusion, and re-ranking configurations within the same RAG pipeline, where these configurations are compared both from the retrieval perspective and from the generation perspective together. System developers building their RAG systems have configuration options available to them but no empirical evidence at generation level to guide their choice. Therefore, configuration options are chosen on the basis of retrieval metrics that, as was shown by Salemi and Zamani (2024), exhibit poor correlation with the factual accuracy of the generated answer.

This research fills this particular gap. With the help of implementation of four retrieval configurations within one controlled RAG pipeline where all generation parameters are kept constant for all the configurations and both perspectives – retrieval and generation – are used for the evaluation, this research provides the necessary empirical evidence that will establish a connection between configuration options and generation-level factual accuracy outcomes. The configurations, evaluation approach and implementation are discussed in Chapter Three.

CHAPTER THREE

Methodology

3.1 Research Design

This study utilizes a controlled experiment research design that helps in isolating the influence of retrieval configuration on the accuracy of answers generated within a Retrieval-Augmented Generation (RAG) pipeline. This research design operationalizes all the research objectives and research questions set out in Chapter One, including comparison of dense only, dense with re-ranking, hybrid fusion, and hybrid fusion with re-ranking in an equal experimental setup.

The independent variable used in this study is the retrieval configuration. Four retrieval configurations are formed in which each retrieval configuration adds one more aspect to the previous retrieval configuration. Dense retrieval serves as the baseline configuration in this study while re-ranking is introduced in the second configuration; then comes the third configuration, where sparse and dense hybrid fusion via Reciprocal Rank Fusion (RRF) is incorporated, while in the fourth configuration, there is hybrid fusion together with re-ranking. This incremental structure allows the individual and combined contributions of fusion and re-ranking to be isolated and compared, directly addressing RQ1 through RQ3.

In order to be sure that any difference in the output quality observed would be due only to the effects of retrieval configuration, a few aspects of the pipeline remained the same for all four configurations. In particular, the generative model, the prompt template, generation settings (temperature, max. token length), the number of context passages passed to the generative model (top-5), the document corpus used, and the evaluation queries were the same for all four configurations. The only aspect allowed to vary between the four configurations was the retrieval and re-ranking system used for retrieving top-5 passages before generation.

All four configurations were evaluated across two levels of assessment: retrieval quality and generation quality. Generation quality was assessed using three related measures: refusal rate, measuring how often the generator declined to answer due to insufficient context; answer correctness (Exact Match and F1), computed against gold-standard answers, evaluating whether the generated answers were factually correct; and Faithfulness, evaluating the extent to which

generated answers were grounded in the retrieved context. Retrieval-level metrics (Recall@5, MRR@5, and Normalised Discounted Cumulative Gain) evaluated the quality of the retrieved passages independently of generation. Faithfulness, from the RAGAS framework, evaluated the extent to which generated answers were grounded in the retrieved context. This multi-level procedure represents the core of the claim made in this research, as discussed in Chapter One, that the quality of retrieval ranking is distinct from the factual correctness and groundedness of generated answers, and that each must be evaluated independently (Salemi & Zamani, 2024).

The entire experimental process was conducted using the Python programming language, where all retrieval modules were executed locally while the generative component was accessed via an application programming interface (API). Implementation details are provided in Section 3.6.

3.2 Dataset

3.2.1 Dataset Selection

The Natural Questions (NQ) benchmark (Kwiatkowski et al., 2019), adapted through the BEIR (Benchmarking Information Retrieval) framework (Thakur et al., 2021), has been adopted for this research study. BEIR Natural Questions is one of the most popular benchmarks for open-domain question answering that includes the actual questions posed by users of Google Search, linked with Wikipedia passages and relevance judgements (qrels). The selection of BEIR NQ benchmark for this experiment has been guided by three criteria set forth in Chapter One, which are as follows; it is a standard benchmark in RAG evaluation literature, it offers pre-existing relevance judgements that help calculate retrieval-based metrics, and its scale and structure support both dense and sparse retrieval paradigms.

3.2.2 Corpus Subset

BEIR Natural Questions benchmark consists of around 2.68 million Wikipedia passages. The entire corpus could not be encoded for dense retrieval due to computational complexity and limitations of available resources, taking up approximately 50+ hours of embedding computation. Therefore, a subset of the first 100,000 passages from the corpus was taken for the

experiment, and the same subset was used to build both the dense (FAISS) and sparse (BM25) indexes to ensure both retrievers operated over an identical document collection (corpus).

The size of subsets chosen for this study is the right compromise between practicality and representativeness. Having 100,000 texts, the corpus remains considerably larger than the small-scale RAG evaluation corpora that are commonly used and allows for generating dense embeddings, creating a FAISS index, and BM25 indexing within one session of computations. The corpus subset remained unchanged in all four setups in order to rule out any differences in the results due to differences in the collection of searchable documents.

3.2.3 Query Sample

Five hundred questions (queries) were randomly selected out of the 3,452 questions in the BEIR Natural Questions test set, with a fixed random seed used (seed = 42), to ensure reproducibility. The same five hundred questions were applied to all four of the retrieval configurations, with the questions' IDs being saved on disk and subsequently loaded at the start of the process for each retrieval configuration in order to make sure the samples used are identical. This makes sure there is no variance due to different question samples when comparing retrieval configurations, thus making performance variances solely due to retrieval configuration differences. Post hoc coverage analysis revealed that 425 out of 500 questions selected (85 percent) have at least one gold relevant passage among the 100,000 documents subset, while the other 75 questions (15 percent) did not have any retrievable relevant passage by design. The 500 questions were kept for evaluation in order to maintain the integrity of the fixed query set, which is considered as a factor for the absolute metric values in Chapter Four.

3.2.4 Relevance Judgments

For the sampled queries, relevance judgments (qrels) were available from the BEIR Natural Questions qrels file, made available by the `ir_datasets` library. Qrels are relevance annotations that match queries with passages deemed relevant to the query by human annotators and which are only used to calculate retrieval-level metrics (Recall@5, MRR@5, NDCG@5), outlined in Section 3.5.1.

3.3 Retrieval Configurations

Four retrieval configurations were created where each configuration corresponds to a unique retrieval strategy commonly used in RAG models. The configurations were created in a progressive manner where each new configuration adds another retrieval improvement to the preceding one, allowing for the isolation of the effects of hybrid fusion and re-ranking individually and in combination. All the four configurations use the same document corpus (Section 3.2.2) for passage retrieval of five passages per query for generation and feed them to the same generator (Section 3.4).

3.3.1 Configuration 1: Dense-Only Retrieval

Configuration 1 is the baseline model for retrieval in this experiment and was used as a benchmark against which all other configurations were evaluated. Configuration 1 implements only dense passage retrieval with no additional re-ranking or fusion.

Dense retrieval was done using Dense Passage Retrieval (DPR). Dense Passage Retrieval consists of a dual encoder architecture that is based on two independently trained BERT-based encoders: a context encoder for the corpus and question encoder for the queries (Karpukhin et al., 2020). The dual encoder structure is key to DPR since the two encoders are trained together but have different embedding functions and, thus, queries and passages have to be encoded by their own encoder for similarity to hold. Passages were encoded by the context encoder (facebook-dpr-ctx_encoder-single-nq-base), while queries were encoded by the question encoder (facebook-dpr-question_encoder-single-nq-base). Both encoders are versions trained on the single dataset – Natural Questions, aligning the retriever with the query distribution used in this study.

For all 100,000 passages in the corpus, the embedding was done using the context encoder into a 768-dimensional dense vector space and these vectors were indexed using FAISS (Facebook AI Similarity Search) using IndexFlatIP – inner-product flat index, which performs exhaustive

search and ensures the exact nearest-neighbour retrieval. The similarity between a query and a passage was calculated as the raw inner-product of their corresponding vectors, similar to the training objective of DPR, which optimises the inner-product similarity of pairs of questions and passages and not the cosine similarity (Karpukhin et al., 2020). The embeddings were therefore not normalised using the L2 norm prior to indexing or retrieval process as the normalisation of the embeddings converts the inner-product similarity to cosine similarity and deviates from model's the scoring function learned during the training.

At query time, for each query, the embedding was generated using the question encoder and the top five relevant passages based on the inner-product similarity was retrieved from the FAISS index. These passages were fed to the generative model without any further processing. This is the simplest retrieval configuration considered in this project.

3.3.2 Configuration 2: Dense Retrieval with Re-ranking

Configuration 2 introduces the extra step of re-ranking to the dense retriever baseline, thus evaluating the isolated contribution of the re-ranking to the overall quality of retrieval and generation separately from any fusion strategy.

In Configuration 2, queries are first encoded and the top fifty candidate passages are identified using the DPR encoder and the FAISS index just like in Configuration 1. The difference is that in Configuration 2 fifty, not five candidate passages were selected initially for re-ranking. This wider initial candidate pool was retrieved specifically to provide the re-ranking model with a sufficient set of candidates from which to identify the most relevant five.

Next, the fifty passages obtained from the first stage were re-ranked based on scores generated using a cross-encoder re-ranker model (cross-encoder/ms-marco-MiniLM-L-6-v2). While the dense retriever utilizes bi-encoders for encoding passages and queries independently and matching them based on vector similarities, a cross-encoder re-ranker encodes both the query and the candidate passage together as a single input. According to Nogueira and Cho (2019), this BERT-based cross-encoder re-ranker achieved state-of-the-art results on the MS MARCO passage ranking task, beating the previous system by 27% relative MRR@10. This shows the superiority of the cross-encoder re-ranker compared to other neural and lexical ranking systems.

The top five passages among the fifty passages obtained from the dense retriever were selected and fed into the generator for answer generation. No fusion of sparse retrieval was used in this configuration, and therefore any difference in performance between Configuration 1 and Configuration 2 is solely because of re-ranking.

3.3.3 Configuration 3: Hybrid Fusion Retrieval

Configuration 3 introduces the use of sparse-dense hybrid retrieval that makes use of the strength of both lexical and semantic retrieval without performing any re-ranking. The purpose of this configuration is to investigate the impact of fusion without re-ranking and, hence, compare it to Configuration 1 and Configuration 2.

Two retrieval approaches were executed independently for each query. Dense retrieval was conducted in the same way as in Configuration 1 through encoding of the query via DPR question encoder and retrieving top fifty passages using inner-product similarity in the FAISS index. At the same time, sparse retrieval was conducted using the BM25 ranking function (Robertson & Zaragoza, 2009), which scores passages using probabilistic term-weighting statistics rather than learned semantic embeddings. BM25 ranking was done over the same 100,000 passage corpus via `rank_bm25` library.

The two ranked lists were combined using Reciprocal Rank Fusion (RRF), a rank-based fusion method that combines several ranked lists by adding up the reciprocal of the rank position of each document in the ranked lists, rather than by adding the raw similarity scores (Cormack, Clarke, & Buettcher, 2009). The RRF value of a document d in ranked list r is computed as $1 / (k + \text{rank}(d))$, where k is a fixed constant. In this experiment, k was set to 60, same as in the original RRF algorithm, where $k = 60$ was found to be close to optimal in some pilot experiments, although it was not found to affect the results significantly. When a document was present in both the dense and sparse ranked lists, then the RRF values for both lists were added up, thus generating a single list that favors documents that are consistently retrieved through both retrieval approaches while also taking into account documents that are retrieved effectively through one approach only.

Rank-based fusion was used instead of score-based combination methods since dense similarity scores and BM25 scores have different numeric ranges and therefore cannot be directly combined without additional normalisation. RRF avoids this problem entirely by operating on rank position rather than raw score magnitude.

After fusion, the top five passages from the ranking produced in the above process were used for generation. Here no re-ranking is done; thus, whatever difference in performance that exists between Configuration 3 and the previous configurations can be solely due to the use of hybrid fusion in this step.

3.3.4 Configuration 4: Hybrid Fusion with Re-ranking

Configuration 4 is the most comprehensive retrieval configuration in this study, as it uses the technique of sparse-dense hybrid fusion along with cross-encoder re-ranking. Thus, it answers RQ3 by checking if the combination of the two techniques results in a more improvement in factual accuracy compared to the independent use of those two techniques.

In this case, just like the previous configuration, the passage retrieval was performed in exactly the same manner; that is, dense retrieval using the DPR encoder and index FAISS and sparse retrieval using BM25 was performed separately to get the top fifty passages using both ranking techniques. These two ranked lists were combined using Reciprocal Rank Fusion exactly as described in Section 3.3.3, with the same $k=60$ constant, producing a single fused ranking of candidate passages.

Unlike in Configuration 3 where the top five fused candidates are sent directly to the generative model, in this configuration, the top fifty fused candidates were sent to the same re-ranker cross-encoder model used in Configuration 2 (cross-encoder/ms-marco-MiniLM-L-6-v2), which ranked all fifty fused candidates independently against the query, and the top five were selected for generation.

The marginal benefit of this configuration is, therefore, that it incorporates both retrieval enhancements used in this experiment sequentially; firstly, fusion is done to incorporate the

lexical and semantic search results into a single ranking, and secondly, the resulting rankings are improved through the use of re-ranking based on query-passage interaction modeling. Since Configurations 1 to 4 were incrementally constructed, whereby each successive configuration introduced an additional retrieval enhancement to the previous one, the marginal difference between Configuration 3 and Configuration 4 isolated the marginal benefit of re-ranking after fusion, while the marginal difference between Configuration 2 and Configuration 4 isolated the marginal benefit of fusion before re-ranking. This incremental design allows both the individual and interactive effects of fusion and re-ranking on factual accuracy to be assessed within a single experimental framework.

3.4 Generation Component

For this experiment, a single generative model was employed in all four retrieval configurations such that differences in the performance of the two would be attributed solely to retrieval configuration, not the language modelling capability of the generator.

3.4.1 Model Selection

Llama 3.2 3B Instruct was selected as the generative model for this experiment, using local inference through Ollama. There are three main reasons why this particular model was selected. Firstly, and most importantly, local inference allows one to do away with the need for the third-party API being available. Everything is done offline, using publicly available model weights and completely independent of any commercial provider of an API. Secondly, at 3 billion parameters, Llama 3.2 3B fits easily into the available consumer hardware memory constraints (Apple Silicon, 16GB unified memory), and still produces coherent and context-grounded text generation. Finally, since the research questions of this study focus on the relative importance of the retrieval system configuration rather than generative model capabilities, using a small and publicly available and locally reproducible model is the optimal way to achieve reproducibility as per the scope stated in Chapter One.

3.4.2 Prompt Design

A fixed prompt template was used across all configurations and all queries:

“Answer the question based only on the provided context. Be concise and factual. If the context does not contain the answer, say "I don't know".

Context 1: [retrieved passage 1]

Context 2: [retrieved passage 2]

...

Context 5: [retrieved passage 5]

Question: [query]

Answer:”

The instruction to respond “only based on the provided context” was used in order to restrict the use of the model to grounded generation without the usage of parametric knowledge from pre-training. It is important in the context of factual accuracy and faithfulness (Section 3.5.2) since it explicitly guides the model to produce answers that can be traced back to the retrieved information. The directive to respond “I don’t know” when the context is not sufficient was used in order to minimize hallucination in case of poor context retrieval; in this way, the rate of refusals itself can be used as an informal metric of context retrieval quality.

3.4.3 Generation Parameters

The generation procedure was performed using temperature of 0.0 and, thus, deterministic greedy decoding. This is a methodological decision since the use of temperature equal to zero ensures identical inputs (query and retrieved context) result in identical outputs, thus, removing sampling randomness as a variable factor for comparison of generated answers between different configurations.

The maximum output length was specified at 150 tokens, enough for direct answering of the natural language questions asked in the Natural Questions dataset format, while keeping generation costs and latencies within bounds on the 500-query test set for each of the four configurations.

3.4.4 Consistency Across Configurations

The number of context passages provided to the generator was kept constant at five for all four configurations, irrespective of which retrieval or re-ranking configuration was employed. This guarantees that the generator always gets a fixed amount of context, regardless of the conditions under consideration, leaving passage relevance and quality, depending on the configuration of the retrieval component, as the variable of interest, rather than passage *quantity*.

3.5 Evaluation Metrics

Two evaluation levels are adopted in this study, one assessing the quality of retrieved passages, separate from the quality of generated answers. This is an essential feature of the design outlined in Chapter One: retrieval metrics assess ranking quality, whereas generation metrics assess whether that quality is reflected in the quality of the output, addressing the gap identified by Salemi and Zamani (2024) between retrieval evaluation and downstream generation quality.

3.5.1 Retrieval-Level Metrics

Three retrieval metrics were calculated for each of the configurations based on the relevance judgments (qrels) of BEIR Natural Questions discussed in Section 3.2.4 for all 500 test queries.

Recall@5 score represents the ratio of known relevant passages of a certain query which can be found among the top five passages retrieved. In other words, it indicates the success rate of a retrieval configuration in retrieving relevant passages within the context window used by the generator.

Mean Reciprocal Rank (MRR@5) is the mean of reciprocal ranks of the first relevant document retrieved in the top five passages. In particular, this score is very sensitive to the rank

of the retrieved relevant passages, since large language models have been found to assign more importance to earlier context when generating output (Liu et al., 2023).

Normalised Discounted Cumulative Gain (NDCG@5) considers both the quality of ranking and the rank of all relevant passages in the top 5, using a logarithmic discount for contributions made by relevant passages that are of low rank. NDCG offers a more holistic view of the quality of the ranking than either Recall or MRR, as it is influenced by the relative ranking of several relevant passages and not just the inclusion or the appearance of a single relevant passage.

It was necessary to use all three of the following metrics as a set because they evaluate different qualities of retrieval results: presence (Recall), early precision (MRR), and ranked ordering (NDCG). Reporting all three allows retrieval-level differences between configurations to be characterised more precisely than any single metric would permit alone.

3.5.2 Generation-Level Metrics

Quality of generation was evaluated along two separate dimensions –factual accuracy, where generated answers were compared with gold-standard answers, and faithfulness, where generated answers were evaluated based on the degree to which they were grounded in the retrieved context. The two dimensions assess two separate constructs, where an answer can be faithful to the retrieved context but inaccurate from a factual perspective if the retrieved context itself is irrelevant, and vice versa, if the model draws on parametric knowledge rather than the retrieved context. Measuring both allows the study to distinguish these cases.

Answer Correctness (Factual Accuracy). Factual accuracy was evaluated through two standard question answering metrics calculated against gold standard short answers: Exact Match (EM) and token-level F1. Exact Match is a binary metric that scores 1 if the normalized answer generated matches the normalized gold answer exactly and 0 otherwise. Token-level F1 is a metric that reflects the amount of token overlap between the generated answer and the gold answer, and it is the harmonic mean of the precision and recall over shared tokens. It thus allows partial scoring for answers that are almost but not exactly right. Both EM and token-level F1 apply the standard SQuAD answer-normalization procedure before evaluation. In case a question had multiple gold answers, the best score out of all the gold answers was considered. Since Natural Questions short answers vary greatly in their length, from one entity to descriptive

phrase, the token-level F1 is considered the primary answer accuracy metric because Exact Match cannot give any score for correct but non-exactly matching answers; both metrics are reported.

The gold standard answers were obtained through joining the sampled queries from BEIR to the original natural questions dataset. Since the identifiers for BEIR queries are artificial and do not correlate with the natural questions' identifiers, the join was done using exact question text matching and was confirmed to be accurate for all the 500 sampled queries. Out of these, 395 (79%) queries had human-annotated short answers in Natural Questions while the rest 105 had only the long-form answers and were thus not included in the answer-correctness assessment. Exact Match and F1 were then computed over the 395 queries with available short answers.

Refusal rate. The prompt constrained the generator to answer "I don't know" in cases when the relevant context could not be found in the retrieved context (Section 3.4.2). Refusal rate was measured per configuration as an independent metric and was calculated over all 500 queries. Refusal rate can be seen as an informal measure of retrieval quality since the higher the rate of refusals the worse the configuration finds the relevant context and, therefore, is needed for understanding the answer correctness and faithfulness metrics. Refusals were included in the calculation of the correctness scores and scored against the gold answers in the same way as any other answer, resulting in a perfect match score of zero and an F1 score close to zero due to lack of overlap with the gold answer. The primary Exact Match and F1 scores consequently represent the behaviour of the entire system for all answered and refused questions. Separate scores not including refusals were also calculated for a picture of the answer correctness for the selected subset of queries where the system committed to an answer.

Faithfulness. Faithfulness was evaluated through the RAGAS evaluation framework (Es et al., 2024). This metric breaks down a generated answer into individual claims and checks those claims against the retrieved context through the use of a large language model judge, outputting a number between 0 and 1 that reflects the fraction of claims supported by the context. Faithfulness was calculated using Llama 3.1 8B Instruct as the judge model, run locally via Ollama. A model bigger than the 3 billion parameter generator was chosen to be the judge because of the potential problem of self-assessment bias that could occur if the model would judge its own work. Sentence embedding operations needed for the RAGAS pipeline employed the all-MiniLM-L6-v2 model. Since refusals ("I don't know") do not include verifiable factual

claims, they were filtered out before faithfulness evaluation; hence, faithfulness is a conditional metric, measuring the groundedness of generated answers given that the system committed to an answer.

This judge model (8B) is much closer in capability to the generator (3B) than what is usually considered acceptable for the evaluation of LLM-as-judge evaluation, when a significantly larger judge is normally used. This is the result of the particular trade-off which was made due to the limitations of the completely local pipeline. The judge model in question is the largest one which can fit into the system together with the rest of the pipeline on the available hardware, while still providing a meaningful margin in terms of capability over the generator. This limitation is acknowledged in Section 3.7.

Two candidate judge models with more capability margin than the generator, namely Phi 3 Medium (14B) and Mistral Small 3.1 (24B), were examined as possible replacements for the Llama 3.1 8B Instruct on a pilot set of ten queries prior to faithfulness scoring on the full-scale dataset. Mistral Small 3.1 did not manage to score any of the ten queries within a 300-second timeout on the hardware available and was ruled out for further consideration in this study due to impracticality for the full-scale experiment. Phi 3 Medium managed to give valid scores to five out of ten queries, with failures providing structurally invalid results, which cannot be used for large-scale scoring. Llama 3.1 8B Instruct was chosen as the judge due to its ability to provide valid scores on all but two queries out of 224, which amounts to 99.1% validity of responses, representing the practical ceiling for judge capability achievable within this study's local, hardware-constrained pipeline.

3.5.3 Sampling for Generation-Level Evaluation

The three evaluation levels were calculated using different query sets, depending on data and computational limitations. The retrieval-level evaluations (Recall@5, MRR@5, NDCG@5) and the refusal rate were calculated using the entire query set of 500 queries. The answer-correctness metrics (EM, F1) were calculated over 395 queries that have a gold short answer in Natural Questions; the 105 queries without gold short answers were excluded from answer-correctness evaluation and contribute only to the retrieval and refusal metrics. Faithfulness was calculated

over a randomly sampled set of 100 queries per configuration with the help of a fixed random seed (seed = 42) which is common to all four configurations. The set of 100 queries per configuration has been saved to disk. Faithfulness sample is significantly smaller compared to retrieval level or answer-correctness evaluations due to the higher costs of calculating RAGAS, where multiple large language model requests are required for claims decomposition and verification against the daily limit of the evaluation infrastructure. Generation-level results based on the smaller faithfulness sample of 100 queries will be discussed accordingly in Section 4.1.

3.5.4 Statistical Analysis

In order to find out if the differences observed between the configurations were statistically significant, pairwise significance tests were performed on all the six configuration pairs using each of the metrics with the scores that were saved for evaluation per-query.

For continuous variables (Recall@5, MRR@5, NDCG@5, token level F1 and faithfulness), the Wilcoxon signed rank test was applied. It is a paired, non-parametric test used to compare two configurations that have been evaluated on the same queries, where normal distribution of score difference is not assumed. It was done in a two-sided manner, with the Wilcoxon zero method for tied pairs. For the binary variables (Exact Match and refusal), McNemar's test with Yates's continuity correction was applied, as it is the appropriate paired test for dichotomous outcomes measured on the same queries under two conditions.

Because six different pairwise comparisons were done within each metric group, correction for multiple comparisons was performed to account for the false discovery rate. The Benjamini-Hochberg procedure was used, with an alpha level of $\alpha = 0.05$, in each of the metric groups within their six pairwise comparisons. Significance is therefore expressed based on the above-adjusted p-values.

For the Wilcoxon signed-rank test, the effect sizes were reported as $r = |Z| / \sqrt{n}$, following the standard normal-approximation effect size for the Wilcoxon signed-rank test.. In this equation, Z is the normal approximation of the test statistic while n is the number of non-zero paired differences. Where the number of non-zero paired differences was too small to compute the test statistic reliably, no result is reported for that comparison; this occurred for the faithfulness

metric, whose scores clustered near the ceiling value and produced few non-zero differences between the configurations (Section 4.1.5).

3.5.5 Confidence Intervals

In addition to the significance tests, 95% confidence intervals were also obtained for all the metrics reported, both for the estimates of each configuration individually, and for each of the pairwise differences between configurations. Confidence intervals for the continuous metrics (Recall@5, MRR@5, NDCG@5, F1, Faithfulness) were calculated using paired bootstrap resampling (2,000 resamples, percentile method, seed = 42). Confidence intervals for individual proportions for the binary metrics (Exact Match, Refusal) were calculated using the Wilson score method; confidence intervals for paired differences were obtained using the Newcombe’s method, which is the paired extension of the Wilson score confidence interval for proportions, consistent with the paired design used for McNemar’s test.

The confidence intervals obtained together with the pairwise comparisons are unadjusted single-comparisons intervals that need to be interpreted together with, not instead of, the Benjamini-Hochberg corrected significance results presented in Section 3.5.4. A few of the comparisons (e.g., Exact Match, Configuration 1 versus Configuration 3) show a confidence interval excluding zero while maintaining non-significance statistically after correction; this portrays the more conservative standard applied once multiple comparisons are corrected for, rather than an inconsistency in the data.

3.6 Implementation Details

This subsection details the software environment, libraries, and reproducibility measures employed in implementing the experimental pipeline.

3.6.1 Computational Environment

The experimental pipeline was written in Python 3.13 using a virtual environment to ensure reproducibility. All local computations, namely the encoding of the documents, construction of the FAISS index, BM25 indexing, and reranking, were done on an Apple Silicon (M4 Pro) device, featuring 16GB of unified memory. Since the cost of encoding all 100,000 documents in

the passage document database was too high, the process of generating the dense embeddings and the FAISS index was done in a cloud-based environment utilizing a GPU (Google Colab, NVIDIA T4 GPU). The generated index was saved to disk and used in all retrieval and generation experiments locally. Such design decisions are typical for the design of a retrieval augmentation generation (RAG) model and do not impact the validity of comparisons between the different configurations at query time, as all configurations used the same pre-built index. Generation and faithfulness judging were performed locally on the same Apple Silicon device using Ollama.

3.6.2 Software Libraries

Component	Library	Purpose
Dense retrieval	sentence-transformers, faiss-cpu	Document/query encoding, similarity search
Sparse retrieval	rank_bm25	BM25 indexing and scoring
Re-ranking	sentence-transformers (CrossEncoder)	Cross-encoder passage scoring
Generation	ollama	Local inference for Llama 3.2 3B Instruct (generator) and Llama 3.1 8B Instruct (RAGAS judge)
Generation evaluation	ragas	Faithfulness metric computation
Answer correctness	<i>(standard library / custom scoring)</i>	Exact Match and token-level F1 against gold-standard answers, using SQuAD-standard answer normalisation

Gold-answer join	datasets	Text-match join between BEIR NQ queries and Natural Questions gold short answers
Statistical testing	scipy, statsmodels	Wilcoxon signed-rank and McNemar's tests; Benjamini-Hochberg FDR correction
Dataset access	datasets, ir_datasets	BEIR Natural Questions corpus, queries, and qrels
Statistical testing	scipy, statsmodels	Wilcoxon signed-rank and McNemar's tests; Benjamini-Hochberg FDR correction; bootstrap and Wilson score confidence intervals

3.6.3 Reproducibility Measures

Several measures were taken to ensure the experimental pipeline produces consistent, reproducible results if executed again:

A constant random seed (seed = 42) was used for all stochastic procedures such as sampling of the corpus, queries, and subsampling for RAGAS evaluation. The same 500 query identifiers were sampled once and saved to disk, then loaded in the same way for all four configuration scripts, guaranteeing that every configuration was evaluated on exactly the same query set. Similarly, an identical set of 100 queries was sampled once and used in Faithfulness evaluation for all configurations (section 3.5.3).

The indexes of retrieval (FAISS dense index and BM25 sparse index) were created once from the constant set of 100,000 documents and saved to disk, making sure that all four configurations had the same constant document collection.

Generation parameters such as temperature (temperature = 0.0), maximum tokens (maximum tokens = 150) were kept constant throughout all configurations, and the same prompt template was used for all four configurations, as specified in section 3.4.

3.6.4 Fault Tolerance

Considering that the generation would depend on a third-party API, along with rate limitations and occasional connectivity issues, all the generated scripts contained checkpointing mechanisms to save progress periodically throughout the execution process onto disk. This way, in case of any disturbance, the process would resume from the last checkpoint, not having to restart the entire set of queries, ensuring computational efficiency and reproducible recovery from transient failures.

3.7 Methodological Limitations

While the experimental design described in this chapter allows a controlled comparison of retrieval configurations, several limitations with the methodology should be acknowledged, consistent with the scope established in Chapter One. These limitations are grouped below according to whether they primarily affect the internal validity of the comparative findings, that is, confidence that observed differences between configurations reflect real effects of retrieval configuration rather than confounds, or the generalisability of those findings beyond the specific conditions tested in this study.

3.7.1 Limitations Affecting Internal Validity

Faithfulness sample and judge. Because of the resource requirements of RAGAS evaluation, a fixed subset of 100 queries per configuration was used to compute the faithfulness score. The judge model used (Llama 3.1 8B Instruct) is closer in capability to the generator (Llama 3.2 3B Instruct) than is typically recommended for LLM-as-judge evaluation, where usually a substantially larger judge is preferred to reliably detect subtly ungrounded claims. This reflects a deliberate trade-off within the resource-limited environment of the fully local evaluation pipeline; the judge is the largest model that could be executed reliably on the available hardware while retaining a meaningful capability margin over the generator (Section 3.5.2). Large-language-model-based evaluation cannot be taken as an exact surrogate for human

assessment of groundedness, and faithfulness scores should be interpreted in light of this. Faithfulness scores were also calculated conditionally, excluding refusals, and therefore reflect groundedness given that the system committed to an answer, rather than across all queries. Because this limitation concerns the reliability of the measurement instrument itself, applied identically across all four configurations, it bears on how confidently the faithfulness *null result* can be interpreted, rather than on whether the finding generalises beyond this study.

Query sample size. Retrieval-level metrics were calculated using a fixed sample of 500 queries, rather than the full Natural Questions test set. This sample size was chosen to balance reliability of metric estimation against computational feasibility, and the same sample was used identically across all four configurations. Despite the fixed sample size, the study retained sufficient statistical power to detect meaningful effects where they existed: five of six pairwise comparisons on each retrieval metric reached significance with small-to-large effect sizes ($r = 0.17\text{--}0.56$, Section 4.1.2), indicating that the consistent null results observed at the generation level are unlikely to be solely an artefact of insufficient sample size.

Answer-correctness coverage. Answer correctness was calculated on the 395 of 500 queries for which Natural Questions provides gold-standard short answers; the remaining 105 queries, which have only long-form answers, were excluded from EM and F1 calculation. While this remains a substantial sample, this exclusion narrows the evaluation for answer correctness specifically relative to the full 500-query retrieval evaluation.

3.7.2 Limitations Affecting Generalisability

Corpus subset. This study evaluates the retrieval configurations using a subset of BEIR Natural Questions consisting of 100,000 passages, compared with the full corpus of approximately 2.68 million passages, a choice necessitated by the hardware resources available for this research (Section 3.2.2). Although this subset is substantially larger than those used in many small-scale RAG evaluations, and was held identical across all four configurations, the absolute retrieval results reported in Chapter Four do not generalise directly to retrieval over the full corpus, where a larger candidate pool could affect the relative performance of dense, sparse, and hybrid retrieval differently. A post-hoc coverage analysis established that 85% of the 500 evaluation queries had at least one relevant passage within the 100,000-document subset, with the

remaining 15% structurally unanswerable regardless of retrieval configuration. This partially explains the relatively low absolute retrieval metric values reported in Chapter Four, though it does not affect the internal validity of the comparative findings, as all four configurations were evaluated under identical corpus conditions.

In-domain retriever. The dense retriever used in this study (DPR) was trained on Natural Questions, the same dataset used for evaluation. Dense retrieval therefore operates in-domain, on its training distribution, representing a best-case scenario for dense retrieval relative to sparse and hybrid alternatives. The benefits of hybrid retrieval documented in the literature tend to emerge specifically where dense retrievers are weaker or operate out-of-domain. The findings of this study regarding hybrid fusion should therefore be interpreted as holding for the in-domain case tested here, and do not generalise to a general claim about hybrid retrieval's value across all deployment conditions.

Single re-ranker model. Consistent with the scope outlined in Chapter One, this study evaluates re-ranking as a configuration choice using a single pre-trained cross-encoder, rather than comparing re-ranker architectures against one another, a question addressed extensively elsewhere (Abdallah et al., 2025; Dong et al., 2024). The findings regarding re-ranking's benefits are therefore specific to this re-ranker architecture and may not generalise to all re-ranking approaches.

Single generative model. The generative model was held fixed across all configurations to isolate the effect of retrieval configuration. Consequently, this study cannot establish whether the relative performance of the four configurations would generalise to generative models of different scale or architecture, which may exhibit different sensitivity to retrieval quality and context composition.

Single RAGAS metric. While the RAGAS framework provides multiple generation-level metrics, this study reports Faithfulness only. Answer Relevancy and Context Precision were not computed, due to API compatibility constraints and the computational cost of running multiple RAGAS metrics across four configurations. Factual correctness is assessed independently in this study through Exact Match and F1 against gold answers (Section 3.5.2), rather than through Faithfulness, which assesses groundedness rather than correctness. The findings of this study are

therefore bounded by the two generation-level constructs actually measured, and do not generalise to the additional dimensions RAGAS provides but which were not computed here.

CHAPTER FOUR

Results Discussion

4.1 Results

4.1.1 Overview of Results

This section presents the empirical results of the four retrieval configurations discussed in Chapter Three. Metrics are reported for two levels of evaluation: retrieval-level metrics (Recall@5, MRR@5, NDCG@5) calculated on the whole 500-query evaluation set; generation-level: answer-correctness metrics (Exact Match and token-level F1) calculated on the 395 queries with gold standard short answers; faithfulness calculated on a fixed 100-query set using the RAGAS framework with Llama 3.1 8B Instruct judge; and refusal; refusal rate is also reported for all 500 queries. All four configurations were tested under identical conditions in terms of the document corpus used, the set of queries, the generative model, and generation parameters, making retrieval configuration the only independent variable.

All pairwise comparisons between the configurations were statistically tested according to the methods described in Section 3.5.4 (Wilcoxon signed-rank for continuous metrics and McNemar's test for binary metrics, Benjamini-Hochberg correction across each metric family). The results will be reported descriptively in this chapter, with discussion in the context of the research questions and previous literature left for Section 4.2.

4.1.2 Retrieval-Level Results

Table 4.1 presents the three retrieval-level metrics for each configuration, computed over the full 500-query evaluation set.

Table 4.1: Retrieval-Level Performance by Configuration (n = 500), with 95% CI

Configuration	Recall@5 [95% CI]	MRR@5 [95% CI]	NDCG@5 [95% CI]
Config 1 Dense Only	0.5518 [0.509, 0.594]	0.4437 [0.405, 0.484]	0.4530 [0.416, 0.491]
Config 2 Dense + Re-ranking	0.6163 [0.576, 0.657]	0.5187 [0.479, 0.558]	0.5281 [0.489, 0.566]
Config 3 Hybrid Fusion	0.5100 [0.469, 0.551]	0.4063 [0.368, 0.446]	0.4151 [0.380, 0.452]
Config 4 Hybrid Fusion + Re-ranking	0.6137 [0.572, 0.655]	0.5154 [0.476, 0.554]	0.5259 [0.487, 0.563]

There are three emerging trends based on the results at the retrieval level, and all of them are significant after corrections for multiple comparisons.

The first trend is that re-ranking significantly enhances all three retrieval metrics in both retrieval approaches. The application of re-ranking to dense retrieval (Configuration 1 → Configuration 2) leads to the enhancement of Recall@5 from 0.5518 to 0.6163 (Wilcoxon, adjusted $p = 0.0006$, $r = 0.35$), and similar significant improvements in MRR@5 and NDCG@5. The application of re-ranking to hybrid fusion (Configuration 3 → Configuration 4) results in an even greater increase in the same metrics, from 0.5100 to 0.6137 (adjusted $p < 0.001$, $r = 0.54$), again with significant gains across all three metrics.

Second, and contrary to the conventional expectation that hybrid retrieval is better than the use of single-method retrieval, hybrid fusion without re-ranking significantly degrades retrieval performance compared to dense retrieval only. The Configuration 3 (hybrid fusion) demonstrates

significantly lower performance than Configuration 1 (dense only) in terms of all three metrics: Recall@5: 0.5100 vs 0.5518 (adjusted $p = 0.0208$, $r = 0.23$); MRR@5 and NDCG@5 have similarly significant differences. The addition of BM25 sparse retrieval to dense retrieval, combined via Reciprocal Rank Fusion, degraded rather than improved the quality of the top five retrieved passages.

Third, when re-ranking was applied, the difference between hybrid and dense retrieval disappeared entirely. Configuration 2 (dense + re-ranking) and Configuration 4 (hybrid fusion + re-ranking) were statistically indistinguishable on all three retrieval metrics (Recall@5: 0.6163 vs 0.6137, adjusted $p = 0.76$). The cross-encoder re-ranker, operating over a pool of fifty candidates, produced an equivalent final ranking regardless of whether those candidates were supplied by dense retrieval alone or by hybrid fusion.

The 95% confidence intervals for each of the pairwise differences in retrieval performance are in line with the significance results reported above; five of the six pairwise comparisons show 95% confidence intervals excluding zero on all three retrieval metrics: Recall@5, MRR@5, and NDCG@5 (for example, the Configuration 1 versus Configuration 2 Recall@5 difference, 95% CI $[-0.099, -0.032]$), while the Configuration 2 versus Configuration 4 comparison, found statistically non-significant on all three metrics, shows a 95% confidence interval that includes zero throughout (Recall@5 difference, 95% CI $[-0.013, 0.019]$).

Taken together, the retrieval-level results indicate that re-ranking was the decisive factor in retrieval quality, that fusion was actively harmful in the absence of re-ranking, and that fusion contributed nothing detectable when re-ranking was present. The two re-ranked configurations were statistically indistinguishable from one another and significantly superior to both non-re-ranked configurations.

4.1.3 Refusal Rates

The generator was instructed to respond "I don't know" when the retrieved context did not contain the answer (Section 3.4.2). Table 4.2 reports the refusal rate for each configuration over all 500 queries.

Table 4.2: Refusal Rate by Configuration (n = 500)

Configuration	Refusal Rate
Config 1 Dense Only	0.422
Config 2 Dense + Re-ranking	0.386
Config 3 Hybrid Fusion	0.476
Config 4 Hybrid Fusion + Re-ranking	0.392

In contrast to the retrieval-level results, the refusal rate did not differentiate between re-ranking and non-re-ranking configurations. Specifically, Configuration 3 (hybrid fusion, no re-ranking) generated the highest refusal rate across all configurations (0.476), which was significantly higher than Configuration 1 (0.422; McNemar, adjusted $p = 0.014$) and Configuration 2 (0.386; adjusted $p = 0.0001$). On the other hand, Configuration 4 (hybrid fusion, re-ranking) showed a

lower refusal rate than Configuration 3 (0.392 vs. 0.476; adjusted $p = 0.0001$), which demonstrates a significant decrease in the number of refusals when re-ranking is performed after the fusion process. At the same time, adding re-ranking to the dense retrieval process led to a smaller decrease that did not reach significance after corrections (0.422 vs. 0.386; adjusted $p = 0.100$) and the latter configuration (0.386) was statistically indistinguishable from Configuration 4 (0.392; adjusted $p = 0.771$). Thus, it can be assumed that the refusal behaviour is driven mostly by the hybrid fusion without re-ranking having the highest refusal rate.

4.1.4 Answer-Correctness Results

Table 4.3 presents answer-correctness metrics computed over the 395 queries for which Natural Questions provides gold-standard short answers. The primary metrics (EM_all, F1_all) are calculated over all 395 queries, with refusals scored against the gold answers in the standard manner, resulting in an Exact Match of zero and near-zero F1. Results conditioned on non-refusal responses are reported separately.

Table 4.3: Answer-Correctness Performance by Configuration, with 95% Confidence Intervals (n = 395)

Configuration	EM (all) [95% CI]	F1 (all) [95% CI]	EM (non-refusal)	F1 (non-refusal)
Config 1 Dense Only	0.1392 [0.109, 0.177]	0.2736 [0.240, 0.310]	0.2350	0.4546
Config 2 Dense + Re-ranking	0.1241 [0.095, 0.160]	0.2766 [0.244, 0.311]	0.2000	0.4335

Config 3 Hybrid Fusion	0.1063 [0.080, 0.141]	0.2470 [0.216, 0.280]	0.1944	0.4396
Config 4 Hybrid Fusion + Re-ranking	0.1266 [0.097, 0.163]	0.2785 [0.244, 0.314]	0.2024	0.4343

However, while there were differences at the retrieval level, none of the configurations performed significantly better than any other with regard to answers correctness. With regard to the main metric F1_all, all six paired comparisons were non-significant (Wilcoxon, all adjusted $p \geq 0.09$) and scores fell into a narrow range between 0.2470 and 0.2785. The same holds true for the Exact Match metric, where all the paired comparisons were non-significant (McNemar, all adjusted $p \geq 0.25$) and scores ranged between 0.1063 and 0.1392. The retrieval-level and refusal-rate improvements produced by re-ranking did not translate into any measurable improvement in the factual correctness of generated answers.

The non-refusal columns are provided for descriptive purposes. They cannot be used for comparison between configurations, since the set of questions for each configuration's non-refusal subset is different: if a configuration refuses to answer on different questions, then it is by design being evaluated on a different set of questions, one which is not equivalent to those answered by other configurations. The apparently better performance of Configuration 1 in terms of non-refusal F1 score (0.5720) thus does not mean that dense-only retrieval provides more accurate answers, but rather that Configuration 1 answered a different set of questions from other configurations. No comparative claim is made on the basis of the non-refusal figures, and no significance testing was performed on them.

4.1.5 Faithfulness Results

Faithfulness was calculated over the fixed 100-query subset using the RAGAS framework with a Llama 3.1 8B Instruct judge, and was scored only on non-refusal responses, as refusals contain

no verifiable claims. Table 4.4 reports the average faithfulness for each configuration together with the full sample accounting.

Table 4.4: Conditional Faithfulness by Configuration, with 95% Confidence Intervals (100-query subset)

Configuration	Non-refusal (n)	Valid scores (n)	NaN (n)	Mean Faithfulness [95% CI]
Config 1 Dense Only	51	51	0	0.8854 [0.817, 0.944]
Config 2 Dense + Re-ranking	63	62	1	0.8368 [0.764, 0.901]
Config 3 Hybrid Fusion	51	51	0	0.8280 [0.746, 0.904]
Config 4 Hybrid Fusion + Re-ranking	59	58	1	0.8291 [0.749, 0.905]

All four configurations scored high conditional faithfulness, with mean scores ranging from 0.8280 to 0.8854 among answered queries. Pairwise differences were not statistically significant. No pairwise comparison reached significance after correction (Wilcoxon, all adjusted $p \geq 0.66$), with effect sizes small to modest across all six pairs ($r = 0.01$ to 0.19). Faithfulness therefore cannot be used to distinguish the configurations and is presented here as a non-comparative result: when a configuration committed to an answer, that answer was in almost all cases well grounded in the retrieved context, regardless of retrieval configuration.

There are two things to keep in mind about the numbers above. The first is that while the judge model (Llama 3.1 8B Instruct) used in the experiments has an arguably smaller capability margin compared to the 3-billion parameter generator, as explained in sections 3.5.2 and 3.7, the faithfulness scores obtained using this model were lower than those close to ceiling values that could be achieved with larger judge models in previous work. This is due to the generally stricter judging behavior of the model. The second is that Natural Questions short answers tend to consist of concise factoids with few distinct factual claims to validate, which might reduce the ability of the metric to distinguish configurations despite the capability of the judge. High faithfulness scores should therefore be interpreted as indicating an absence of detectable unsupported claims, rather than as a fine-grained measure of generation quality.

4.1.6 Summary of Findings

This section has presented the empirical results of the four retrieval configurations across the two levels of evaluation. The findings are summarised in Table 4.5.

Table 4.5: Summary of All Results

Configuration	Retrieval			Generation		
	Recall@5	MRR@5	NDCG@5	Refusal	F1 (all)	Faithfulness

Config 1 Dense Only	0.5518	0.4437	0.4530	0.422	0.2736	0.8854
Config 2 Dense + Re-ranking	0.6163	0.5187	0.5281	0.386	0.2766	0.8368
Config 3 Hybrid Fusion	0.5100	0.4063	0.4151	0.476	0.2470	0.8280
Config 4 Hybrid Fusion + Re-ranking	0.6137	0.5154	0.5259	0.392	0.2785	0.8291

There are three key findings.

First, re-ranking significantly improved retrieval accuracy and significantly decreased the refusal rate, both under dense and hybrid retrieval conditions. Among the three retrieval-level enhancements compared in this study, re-ranking was the only one to produce a statistically significant *improvement*; hybrid fusion, by contrast, produced a statistically significant *degradation* when applied without re-ranking (see Finding 2).

Second, hybrid fusion did not improve performance and significantly worsened retrieval quality compared with dense retrieval alone. When combined with re-ranking, fusion had no measurable effect on retrieval quality compared with dense retrieval with re-ranking, on any metric; this pattern was consistent across all three retrieval measures, with five of six pairwise comparisons reaching significance on Recall@5, MRR@5, and NDCG@5 alike, the sole exception being the Configuration 2 versus Configuration 4 comparison in every case (Section 4.1.2).

Third, and most importantly, none of the retrieval-level or refusal-rate improvements had an effect on the end-to-end correctness of the generated answers. No configuration differed statistically from any other on Exact Match or F1, and faithfulness scores did not differ significantly across configurations. This null result is corroborated by the 95% confidence

intervals computed for each pairwise difference (Section 3.5.5): all six F1 pairwise confidence intervals include zero, and all six Faithfulness pairwise confidence intervals are wide and include zero, consistent with the absence of statistically detectable differences at either metric.

Rather than pointing to a single optimal setup, the findings reveal a split between levels of evaluation: the two re-ranked configurations outperformed the other two on retrieval metrics, refusal behaviour followed a more complex pattern driven primarily by the fusion-without-re-ranking configuration, and none of the configurations could be distinguished from the others in terms of answer correctness or faithfulness. This split between retrieval-level and generation-level outcomes is the main empirical finding of this research project, directly supporting the argument made in Chapter One that retrieval-level and generation-level performance are distinct constructs that do not necessarily correlate with each other. Its interpretation, mechanisms, and implications for RAG system design are examined in Section 4.2.

4.2 Discussion

Having presented the empirical results in Section 4.1, this section turns to their interpretation in relation to the study's research questions and the existing literature.

4.2.1 Introduction

Section 4.1 outlined the results from four different retrieval configurations assessed across two domains: retrieval quality and generation quality. The pattern of results was surprising. At the retrieval level, different configurations gave rise to significant differences which were both

statistically robust and theoretically meaningful: re-ranking helped improve ranking quality, hybrid fusion without re-ranking reduced ranking quality, and refusal behaviour varied across the different configurations. At the generation level, none of these differences translated into measurable improvements in answer correctness or faithfulness.

This section seeks to interpret this pattern with respect to the three research questions formulated in Chapter One, situate the results in the existing literature outlined in Chapter Two, and think about the theoretical and practical consequences of the discrepancy between the results at retrieval level and at generation level. Section 4.2.2 addresses RQ1, the impact of hybrid fusion on factual accuracy. Section 4.2.3 addresses RQ2, the impact of re-ranking. Section 4.2.4 addresses RQ3, the joint impact of both approaches. Section 4.2.5 focuses on the main dilemma emerging from all three sets of results: that of the discrepancy between improvement at retrieval level and the result at generation level. Section 4.2.6 explores practical consequences of this dilemma for the design of RAG systems. Section 4.2.7 discusses the limitations of the study in light of the results and suggests avenues for future work.

4.2.2 The Effect of Hybrid Fusion on Factual Accuracy (RQ1)

The first research question examined whether hybrid fusion retrieval impacts factual accuracy when compared to dense retrieval alone. There is a definitive answer at the retrieval level and a null answer at the generation level, and the difference between the two speaks volumes.

In terms of retrieval performance, Configuration 3, in which hybrid fusion without re-ranking is used, underperforms significantly compared to dense retrieval alone (Configuration 1) by all three metrics: Recall@5, MRR@5, and NDCG@5 (Section 4.1.2). This observation is in contrast with the typical perception in the literature regarding the usefulness of hybrid retrieval that it is a simple extension of retrieval that combines two complementary approaches: lexical matching and semantic matching (Section 2.2.3). This is due to the particular conditions in which dense retrieval was tested in our work. The DPR encoder was trained on Natural Questions, the dataset we use for evaluation (Section 3.7). That means that we have an in-domain case, when the dense retriever works exactly on the query distribution for which it was trained, which corresponds to the most optimal dense retrieval setting compared to sparse and hybrid retrieval. Indeed, most of the literature about the benefits of hybrid retrieval, including the Reciprocal

Rank Fusion (Cormack et al., 2009) paper, and the modern implementations of hybrid retrieval in the RAG pipeline, is based on out-of-domain dense retrieval cases, when the dense retriever is either weak, or more general-purpose, and thus cannot capture certain aspects of the query distribution. In such settings, the precise lexical matching of sparse retrieval can be truly complementary to a dense retriever, which has some kind of blind spots. When the dense retriever is strong and well-matched to the query distribution, BM25 via rank fusion is not helpful in providing additional information but looks like noise, i.e., passages that are lexically relevant but not necessary semantically.

This interpretation is consistent with, although not evidence of, a theory whereby the advantage of fusion depends on the retrievers being fused making truly diverse and complementary mistakes, rather than one retriever doing better than the other consistently, something inherent to the motivation behind the combination of lexical and semantic approaches (Section 2.2.3). This means that the finding reported here should be scoped carefully: this study does not show that hybrid fusion is disadvantageous in general. Rather, it shows that hybrid fusion impaired performance under a specific and well-defined condition, an in-domain, well-trained dense retriever, and this scoping is itself an important and meaningful contribution, since it strengthens rather than contradicts the existing academic literature on hybrid retrieval.

Generation-level results paint a different picture altogether. Neither Exact Match nor F1 scores were significantly different between Configuration 1 and Configuration 3 (Section 4.1.4), nor was faithfulness (Section 4.1.5). The retrieval-level performance drop observed for fusion was significant and meaningful, yet the impact of hybrid fusion on whether generated answers were correct or faithful to context was statistically undetectable. Refusal rates changed, Configuration 3 produced the highest refusal rate of all configurations, significantly higher than Configuration 1 (Section 4.1.3). It seems the impact of hybrid fusion at the retrieval level was enough to make the generator judge that the retrieved context was insufficient to answer from, but not enough, or of the wrong kind, to affect the accuracy of the answers provided. The difference between a configuration influencing the decision of the system to try answering and a configuration affecting the correctness of the answer itself appears again and again through this chapter and is explored in Section 4.2.5 in detail.

Regarding RQ1, then, the answer turns out to be two-fold instead of singular, hybrid fusion significantly degraded retrieval quality and significantly increased refusal rates, but had no statistically detectable influence on the factual accuracy or faithfulness of answers.

4.2.3 The Effect of Re-ranking on Factual Accuracy (RQ2)

The second research question (RQ2) sought to determine to what degree re-ranking could increase downstream factual accuracy with fixed retrieval conditions. Just as with RQ1, the answer varies dramatically based on the evaluation level considered and represents, perhaps, the most significant findings in the entire study.

"Re-ranking was the single most consistently influential factor at the retrieval level in this research. It showed improvements in Recall@5, MRR@5, and NDCG@5 metrics under both retrieval conditions evaluated in the research (dense-only - Configuration 1 vs. Configuration 2 and hybrid fusion - Configuration 3 vs. Configuration 4) with the higher of these two improvements being seen specifically in re-ranking after the fusion process (Section 4.1.2). These results coincide with the theoretical explanation of the re-ranking case given in Section 2.3: a cross-encoder, encoding both the query and each candidate passage in a single process, can capture the interactions between tokens that bi-encoder retriever encoding cannot, thus distinguishing the subtle relevance levels within a candidate pool that the retriever alone could not discriminate. The fact that re-ranking was the most effective on the pool that was created through the fusion process is significant: it means that re-ranking process was able to identify the truly relevant passages in the pool that fusion had made less relevant, thereby correcting for the noise that Section 4.2.2 attributes to fusion in this in-domain setting.

However, refusals exhibited a similar but not an identical trend. Re-ranking yielded a significant decrease in refusals when used following fusion but had no significant impact on refusals when applied under dense-only retrieval (Section 4.1.3). This discrepancy suggests that the effect of re-ranking on refusals is specific to the case when there is a need for correction of the candidate set – i.e., when the candidate set has been created through fusion – rather than providing some universal benefit to the refusal rate.

Nevertheless, at the generation level, re-ranking failed to deliver anything meaningful. Neither Exact Match, F1, nor faithfulness was found to differ significantly between any re-ranked and any non-re-ranked configurations (Sections 4.1.4, 4.1.5). This is one of the clearest examples of the divergence between retrieval and generation in this paper, because re-ranking delivered the most significant and theoretically motivated retrieval-level improvements among all the interventions considered in this work, yet they did not lead to better generation performance.

Two non-exclusively related reasons can be put forward here, the first being based on the difference between what Recall@5 actually measures and what F1 measures. The use of re-ranking in the current work was trained on query-passage relevance (Nogueira & Cho, 2019). This concept of matching means semantic or topical match and is different from finding a clear answer span in a passage. Relevance can be determined by a cross-encoder without the existence of the particular short string required for answering the query. Recall@5 is measured against the relevance of documents from BEIR's corpus, so it rewards topical matches in the same way; F1 is computed against short gold answer strings instead. Thus, it is possible that re-ranking promoted the retrieval of passages that had higher document-level relevance but did not necessarily have an extractable short answer string, a gap invisible to retrieval metrics but directly consequential for answer correctness.

Additionally, the generator used in this work is a comparatively small 3-billion parameter model (Section 3.4.1). It receives instructions of answering briefly or refusing in a concise way. Hence, even in the case of successful retrieval of a passage having the correct answer to the question, answering correctly is a generation-level task that is not performed by retrieval. If the limiting factor in this pipeline was the capability to extract and state the information, and not its presence in context, then no matter how good the improvement of retrieval quality is, the gains will be diminishing or non-existent on correctness, which is indeed the case.

Concerning the RQ2, re-ranking led to significant improvements in retrieval quality and to reduction of refusals in fusion, but not to improvement in accuracy or faithfulness under either retrieval condition tested.

4.2.4 The Combined Effect of Fusion and Re-ranking (RQ3)

RQ3 asked about the relative advantage of combining fusion and re-ranking for gains in factual accuracy compared to their application separately. As with RQ2, there is a discrepancy between the retrieval and generation results, and the specific shape of that divergence adds a further layer to the picture built in Sections 4.2.2 and 4.2.3.

In terms of the retrieval metrics, Configuration 4 (hybrid fusion with re-ranking) had the best scores of all configurations in Recall@5, MRR@5, and NDCG@5 (Section 4.1.2). However, it did not differ statistically from Configuration 2 (dense retrieval with re-ranking) on any of the three metrics. Again, this is quite a different result than the simple "combination is better" statement. It suggests that re-ranking neutralizes the retrieval-level costs of hybrid fusion: even with a cross-encoder re-ranker that worked over fifty candidate passages, there was no measurable difference between a final ranking generated on the basis of fifty candidates obtained through dense retrieval and fifty candidates generated through hybrid fusion described in Section 4.2.2. In other words, not only did re-ranking improve fusion; it cancelled out the retrieval-level costs of fusion altogether, without requiring its presence to arrive at the same level of performance, since Configuration 2 (no fusion, only re-ranking) was statistically identical to Configuration 4 (re-ranking and fusion combined).

This has a direct bearing on how RQ3 should be addressed. Combination of fusion and re-ranking did not lead to a retrieval-level result that would be better than the result obtainable via re-ranking alone; it led to an equivalent result, obtained via a more complex process, because Configuration 4 needed two rounds of retrieval and rank-fusion in addition to re-ranking, whereas Configuration 2 needed only dense retrieval and re-ranking (Section 3.3). Refusal rate supports this interpretation, because the refusal rate of Configuration 4 was statistically identical to the refusal rate of Configuration 2, and both of them were significantly lower than the refusal rate of Configuration 3 (Section 4.1.3), which shows that re-ranking alone was enough to produce the refusal-rate improvement normally attributed to the combination of re-ranking and fusion.

As far as generation level is concerned, the null hypothesis again proved true: there were no differences in Exact Match, F1, and faithfulness between Configuration 4 and any of the other

configurations, including Configuration 2 (Sections 4.1.4, 4.1.5). Thus, using both retrieval enhancements did not lead to any generation-level improvements over re-ranking alone, nor to any generation-level costs as a result of retrieval-level noise introduced by hybrid fusion in Section 4.2.2. Factuality and faithfulness of the system's outputs were, as far as the precision allowed by the samples in this study goes, equally unaffected by the use of fusion or its absence, with re-ranking being applied.

In summary, the answer to RQ3 is that the combination of fusion and re-ranking does not lead to gains in factual accuracy greater than gains resulting from applying re-ranking independently. On the retrieval level, the combination did not excel the results obtainable via re-ranking alone; it merely matched them and incurred additional computational cost. On the generation level, neither the combination nor any individual configuration differed from any other. This finding, combined with those of Sections 4.2.2 and 4.2.3, suggests a consistent practical conclusion elaborated further in Section 4.2.6: under the conditions of this study, re-ranking was the enhancement that matters, and fusion brought no measurable benefits on any level, and incurred retrieval-level costs in case of its use without re-ranking.

4.2.5 Retrieval Metrics versus Generation Metrics: The Central Tension

Each of the research questions posed in Sections 4.2.2 through 4.2.4 yielded the same underlying pattern: retrieval configuration substantially changed retrieval-level metrics, but not a single alteration yielded a statistically significant change in either answer correctness or answer faithfulness. This section takes a step back from the individual research questions to address this pattern directly, which is the central empirical finding of this study.

The pattern is captured succinctly by the results presented in Table 4.5. All retrieval-level metrics distinguished at least some configurations from others, often with large effect sizes (Section 4.1.2). Refusal rate distinguished configurations in accordance with a pattern determined largely by the poor performance of hybrid fusion without re-ranking (Section 4.1.3). But in eighteen pairs of configurations tested for differences in F1 and Exact Match (six pairs $\times$ two metrics, ignoring faithfulness for the time being), not one yielded a statistically significant difference

after adjustment (Section 4.1.4). Likewise, faithfulness failed to yield statistically significant differences in any pairwise comparison (Section 4.1.5).

This is not a vague or tentative finding. It is a specific null result, twice independently replicated; once with factual correctness measured against gold-standard answers and once with groundedness scored by an independent LLM judge, regardless of the differences in retrieval-level metrics preceding it, whether those differences resulted from fusion (Section 4.2.2), re-ranking (Section 4.2.3), or a combination of the two (Section 4.2.4).

This finding is precisely the empirical validation that the research questions in Chapter One called for. Chapter One posited a gap between retrieval-level evaluation and generation-level quality assessment based upon Salemi & Zamani's (2024) argument that traditional retrieval metrics correlate poorly with downstream RAG performance (Section 2.4.1), and Yu et al.'s (2024) description of the relationship between retrieval metrics and generation output as an unresolved research problem (Section 2.4.3). What this study adds to this argument is a controlled experimental demonstration of the gap: four configurations, evaluated under identical generative conditions, showing real and statistically significant differences in retrieval metrics but yielding no statistically distinguishable generation-level differences, regardless of the specifics.

Why is there a gap, mechanically speaking, aside from the reasons given in Sections 4.2.2 and 4.2.3? Three explanations – not necessarily mutually exclusive – may be drawn out:

The first is a construct validity explanation. Recall@5 and its accompanying metrics measure whether a relevant document, as judged by BEIR annotators, appeared in the top five documents returned. F1 and Exact Match measure whether a specific and usually short answer span was retrieved. These are closely related but distinct constructs: a relevant document could appear at the top of the list of search results and yet require inference, synthesis, or careful reading to extract the particular fact needed for the answer. This gap between the metrics of retrieval and the metrics of answer correctness is hidden at the retrieval level and becomes apparent only with

generation-level evaluation, precisely the motivation behind the design of this study's two-level framework (Section 3.1).

The second is a generation capacity explanation. The generator used in this study was deliberately small, 3 billion parameters, and run under a carefully designed and concise prompt (Section 3.4.1). The bottleneck constraining end-to-end correctness is less "Is the right passage present?" and more "Can this generator reliably read and extract the correct short answer from the provided context?". In this case, improvements to retrieval quality, however substantial, will be expected to yield diminishing returns since the bottleneck will have moved downstream. This explanation is consistent with, though by no means conclusive in light of, the pattern of results observed here: substantial and repeated improvements to retrieval metrics, no corresponding generation-level improvement.

The third is a measurement sensitivity explanation, applicable specifically to faithfulness. As explained in Section 4.1.5, faithfulness scores in this study were computed for the non-refusal subset of answers using a model with a relatively small capability advantage over the generator (Section 3.5.2), and Natural Questions' short answers are known for having few distinct facts to check. It is plausible that faithfulness scores, as computed here, simply lacked sufficient statistical power to pick up on any existing generation-level differences, regardless of whether such differences existed.

These three explanations are not mutually exclusive, and this study's design does not allow for their distinction. What can be said with certainty, backed by the significance-tested results in Section 4.1, is the empirical pattern itself: statistically significant retrieval-level differences, which were found and which were substantial, did not propagate to become statistically detectable generation-level differences across two distinct generation-level metrics and every research question posed in this study. Regardless of the precise combination of the underlying mechanisms, the practical implication is clear and is elaborated in Section 4.2.6: retrieval-level metrics alone are not, in this study, a reliable proxy for the generation-level results which ultimately determine the usefulness of the RAG system.

4.2.6 Implications for RAG System Design

The results obtained by this study have important implications for the practical design of RAG systems because they relate to a type of configuration choice faced by such designers. As Section 1.3 explained, these choices are made based on retrieval-level metrics without generation-level data. This section considers what the present findings suggest a practitioner should do differently, and where the findings' scope limits how far that guidance can be generalised.

First of all, re-ranking has been confirmed by the results of this study to be beneficial in every respect. Under both retrieval conditions and across all metrics, it provided statistically significant and frequently very large improvements (Sections 4.1.2, 4.1.3, 4.2.3). Even though these improvements did not translate into better answer correctness in this study, even on its terms, a system which has higher recall and refuses to answer questions less often is, clearly, a better system in practical terms. Therefore, on the basis of the results of this study, a practitioner should consider a re-ranking pass and the additional computational cost it requires, as a generally justifiable option.

Hybrid fusion should not be considered as a default enhancement, especially when dense retrieval is already matched with the deployment domain. The results of this study (Section 4.2.2) have shown that fusion can substantially degrade retrieval quality in case of an in-domain dense retrieval. These results directly undermine the common practitioner approach to fusion as a safe and close-to-strictly-beneficial choice. However, when the dense retriever has been trained or fine-tuned using the data that is closely resembling deployment query distribution, these results suggest that fusion may introduce more noise than additional signal. Nevertheless, this does not mean that fusion should not be used at all. The scoping done in Section 4.2.2 suggests that the benefit from the fusion may depend on the match of the dense retriever and the domain. Therefore, the practical recommendation in this case is not "avoid fusion" but "test the benefits of fusion for the specific dense retriever and domain empirically, rather than assume that it exists by default".

The combination of fusion and re-ranking is not a "belt-and-suspenders" free improvement because it adds computational costs for no measured benefit over re-ranking alone. According to Section 4.2.4, Configuration 4 was statistically indistinguishable from Configuration 2 on every retrieval-level metric. Hence, the additional sparse retrieval pass and fusion step have brought no

measurable benefit in terms of retrieval once re-ranking was added. From the viewpoint of a practitioner evaluating the trade-off between latency and computational budget, which is an important issue in a deployed RAG system, this result implies that when re-ranking is present, there may be no justification for hybrid fusion in terms of computational costs, at least under conditions resembling those tested here.

Perhaps, the most important implication concerns the practice of the evaluation itself, rather than any single configuration choice. The central result of this study – the lack of correlation between substantial and statistically robust improvements in retrieval-level metrics and detectable improvements in generation-level outcome – cautions against the common approach of selecting and tuning RAG configurations based on retrieval metrics alone. A practitioner, relying only on Recall@5 or similar metrics to compare different configurations, would, on the basis of this study's data, easily conclude that Configuration 2 or Configuration 4 is superior, with no clue that this superiority does not correspond to any detectable generation-level improvement. This is exactly the problem highlighted by Salemi and Zamani (2024) in introducing eRAG as an alternative to retrieval-based evaluation based on downstream performance (Section 2.4.1). The results of this study provide direct empirical confirmation of this concern in the specifically designed context. In case of the resources to do so, a practitioner should compare candidate retrieval configurations not based on retrieval-level metrics but on generation-level outcomes and factual correctness against reference answers in case of their availability or human/automated faithfulness assessment otherwise, rather than relying on retrieval-level metrics as a proxy for system quality.

Finally, a more tentative implication concerns the nature of the bottleneck in RAG performance improvements. If, as discussed in Section 4.2.5, the limiting factor for generation-level performance in this study was generation-side extraction capability rather than retrieval quality, then this study suggests that for the systems based on comparably small generators, further investment in retrieval may not pay off unless it is accompanied by corresponding attention to the generator's capacity to use improved context effectively. This implication is proposed tentatively, because the design of this study, with the fixed generator, cannot provide a direct evidence in favor of this statement. This is one of the key directions of further research, outlined in Section 4.2.7.

4.2.7 Limitations and Future Work

The methodological limitations of this study were outlined in Section 3.7, distinguishing those primarily affecting internal validity (3.7.1) from those primarily affecting generalisability (3.7.2). This section identifies the specific directions for future work that follow from each, without repeating the limitations themselves.

Following from the in-domain retriever limitation (3.7.2): the most direct next step is to repeat this experiment using a dense retriever that is general-purpose or genuinely out-of-domain relative to the evaluation corpus, testing whether hybrid fusion's benefit re-emerges, or scales, as domain match weakens. This would convert the scoped claim in Section 4.2.2 into a directly tested one.

Following from the corpus subset limitation (3.7.2): extending this study to the full 2.68-million-passage BEIR Natural Questions corpus would allow the absolute performance figures, though not the core comparative findings, to be assessed under conditions more representative of production-scale deployment.

Following from the single generative model limitation (3.7.2), read together with the generation-capacity explanation raised in Section 4.2.5: repeating this study's four-configuration comparison with a substantially larger generative model, holding all retrieval configurations and evaluation procedures constant, would help determine whether the generation-level null results reflect a capacity bottleneck specific to the small generator used here. A different result under a larger generator would support the generation-capacity explanation; persistence of the null result would instead support the construct-validity explanation, that retrieval-level and generation-level correctness measure genuinely distinct constructs regardless of generator capability.

Following from the faithfulness judge limitation (3.7.1): re-scoring the generated answers with a substantially more capable judge model, should one become practically available, would help establish whether the faithfulness null result persists under a larger capability margin or is partly attributable to the judge itself.

Following from the query sample and answer-correctness coverage limitations (3.7.1): a larger query sample, extending beyond the 395 queries with available gold short answers used here, would increase statistical power to detect smaller effects and strengthen confidence in the null result reported in this study, should it persist.

Following from the single re-ranker architecture limitation (3.7.2), and given the central role of re-ranking in this study's findings (Section 4.2.3): testing whether the retrieval-generation divergence observed here holds across different re-ranker architectures, particularly re-rankers trained with explicit attention to answer-span extractability rather than document-level relevance alone, would establish whether the divergence is a property of re-ranking generally or specific to the architecture used in this study.

Following from the single RAGAS metric limitation (3.7.2): incorporating Answer Relevancy and Context Precision in future work would provide a fuller picture of generation quality and might reveal configuration-dependent differences not visible through the two constructs measured here.

Taken together, these directions represent extensions and stress-tests of the boundaries of this study's findings, not challenges to their internal validity. The comparative findings reported in Section 4.1 and interpreted in this chapter were obtained under a controlled, consistent experimental framework in which all four configurations were evaluated under identical conditions of corpus, query set, generative model, and evaluation methodology.

CHAPTER FIVE

Summary, Conclusions and Recommendations

5.1 Summary of the Study

The purpose of this study is to address an explicit gap in the existing literature on Retrieval-Augmented Generation (RAG), specifically, the lack of systematic empirical evidence on the influence of different retrieval configuration approaches, dense retrieval, hybrid fusion, and re-ranking on not only the quality of ranking of retrieved passages but also the factual accuracy and faithfulness of the answers generated by a RAG system. In accordance with the findings presented in Chapter One, the vast majority of research in this area have examined the effects of various retrieval techniques independently or in pairs, applying solely the retrieval-level metrics of their evaluation, leaving practitioners without generation-level evidence to inform configuration decisions.

In order to close the identified gap, four retrieval configurations were implemented in the single controlled RAG experiment framework: dense-only retrieval, dense retrieval with re-ranking, hybrid fusion retrieval, and hybrid fusion with re-ranking. All four configurations were tested on the same 500-query test set, extracted from the BEIR Natural Questions dataset, using the same document corpus, the same generative model, and the same generation parameters, so that all the differences between configurations' outcomes could be explained exclusively by the variation of the retrieval approach used. The quality of retrieval was measured with Recall@5, MRR@5, and NDCG@5 metrics while the quality of generation was assessed using refusal rate, answer correctness against gold-standard answers (Exact Match and token-level F1), and the level of faithfulness to retrieved context (RAGAS). All the pairwise comparisons of the configurations were tested for the significance of the differences with Wilcoxon signed-rank tests for continuous metrics and McNemar's tests for binary metrics with Benjamini-Hochberg adjustment for the false discovery rate.

5.2 Summary of Findings

The findings of this study, outlined in Sections 4.1 and interpreted in Section 4.2, can be summarized as follows.

At the retrieval level, re-ranking turned out to be the only consistent advantage in configuration settings, providing statistically significant improvement on Recall@5, MRR@5, and NDCG@5 both in dense-only and hybrid fusion setups. The setup with hybrid fusion without re-ranking, however, led to significant worsening of the retrieval compared to dense-only retrieval, what could be explained by the in-domain nature of the dense retriever employed in the current study and serving as the best-case scenario for dense retrievers as opposed to sparse and hybrid ones. After applying re-ranking, the difference between the dense-only and hybrid fusion retrieval configurations vanished completely, suggesting that re-ranking mitigated the retrieval cost of fusion without the need to use fusion itself.

The refusal rate followed a similar yet somewhat different pattern, namely, the highest refusal rate could be observed for the hybrid fusion without re-ranking configuration with the re-ranking significantly lowering the refusal rate only when applied after fusion.

At the generation level, no configuration turned out to differ significantly from any other one in terms of Exact Match, token-level F1, or Faithfulness scores. This null result could be observed for all research questions and two generation-level metrics, namely, factual correctness measured against gold-standard answers and groundedness evaluated by an independent LLM judge.

5.3 Conclusions

In regard to RQ1 (the effect of hybrid fusion on factual accuracy relative to dense retrieval), this study concludes that hybrid fusion substantially degraded the retrieval quality and raised the refusal rate in the in-domain condition used in the study but has no statistically significant effect on factual accuracy or faithfulness. This finding must be viewed in the context of dense retrieval being a well-matched retriever to the evaluation domain but cannot be considered a general statement regarding the usefulness of hybrid retrieval per se.

In regard to RQ2 (the extent to which re-ranking improves downstream factual accuracy), this study concludes that re-ranking results in a substantial and statistically robust increase in retrieval quality and decrease in refusal rate under fusion conditions and does not lead to statistically significant improvement in factual accuracy or faithfulness under either retrieval condition.

In regard to RQ3 (the extent to which the combination of fusion and re-ranking exceeds the performance of each technique individually), this study concludes that the combined techniques did not outperform re-ranking in the retrieval-level measures while incurring an additional computational cost and did not outperform re-ranking alone in the generation-level measures.

In summary, this study shows that statistically robust retrieval-level improvements do not translate into any statistically discernible generation-level improvements in terms of factual accuracy or faithfulness of generated answers under the particular pipeline, generator, and evaluation conditions. This constitutes direct evidence in favor of the key argument put forward in Chapter One: that retrieval-level quality and generation-level factual accuracy are distinct measures and that improvement in one measure does not necessarily result in improvement in the other.

5.4 Contributions of the Study

This study makes four key contributions to the extant body of knowledge on RAG evaluation.

First, it presents a controlled, four-configuration comparative evaluation of dense retrieval, hybrid fusion, and re-ranking within the same RAG pipeline, assessed concurrently at both retrieval and generation levels, a combination which has not been presented before in the relevant literature discussed in Chapter Two.

Second, it proves empirically the hypothesis formulated by Salemi and Zamani (2024) theoretically and listed by Yu et al. (2024) as an open problem, i.e., that retrieval-level metrics have poor correlation with the generation-level results in the framework of RAGs. This study

provides concrete proof of that phenomenon in the form of formal statistical significance testing instead of correlation statistics only.

Third, it presents a limited but mechanistic explanation of why hybrid fusion did not boost but diminished performance in the current study, citing the domain-specific similarity between the retriever and the evaluation corpus as the reason for that particular observation, rather than treating the finding as a general claim about hybrid retrieval's value in a RAG pipeline.

Fourth, it implements a completely local, open-weight, API-independent RAG evaluation pipeline including retrieval, generation, and LLM-based faithfulness judgment modules, providing researchers a way of replicating RAG evaluations without reliance on any commercial API.

5.5 Recommendations

For practitioners designing RAG systems:

In most cases, re-ranking can be considered a justified addition to a retrieval pipeline based on the consistency and magnitude of retrieval-level improvements observed and the role of re-ranking in decreasing the refusal rate, even if the improvements in generation-level correctness are not guaranteed.

The adoption of hybrid fusion as an automatic improvement in general cannot be considered a good idea until empirical validation is performed for a particular dense retriever and particular domain because, in this case, the application of fusion turned out to be detrimental for dense retrieval.

If the computational cost is a limitation, the combination of hybrid fusion with re-ranking cannot be expected to be better than only re-ranking in terms of retrieval level performance; in this experiment, no statistical differences were detected between these two approaches.

For the evaluation of potential retrieval configurations, if practitioners have enough resources, they should use generation-level measures, factual correctness compared to references, or faithfulness of generations, rather than relying only on retrieval-level measures as their substitutes for assessing system quality.

Recommendations for future research:

Future research should repeat this study's experimental design using a dense retriever which is general-purpose or out-of-domain relative to the evaluation corpus in order to check if the benefit of hybrid fusion appears under conditions where it is less beneficial for dense retrieval compared to this experiment.

In future work, this study's comparison should be replicated using a larger generative model in order to investigate if the null results on the generation level presented in this study are caused by some limitations on generation capacity of a smaller model used in this experiment or they reflect a more fundamental distinction between retrieval-level and generation-level concepts that would persist regardless of generator capability.

In future work, this experiment should be expanded to the whole BEIR Natural Questions corpus, to a larger sample of queries, and to other generation-level metrics, including Answer Relevancy and Context Precision, in order to check if the results obtained in this study remain valid in the conditions of higher statistical power and more broad construct coverage.

In future work, the impact of re-rankers' architectures trained explicitly to extract answer spans should be explored as an alternative approach to obtain retrieval-level improvements that could propagate to generation-level correctness more reliably than a general-purpose cross-encoder used in this study.

5.6 Concluding Remarks

This study aimed at establishing the impact of improvement in retrieval configuration on the factual accuracy of RAG-generated answers. From the findings gathered in this research, the answer that emerges is very clear: it does not happen reliably, and it is not automatic either. Configuration 4 (hybrid fusion with re-ranking) achieved the highest scores of any configuration on every retrieval-level metric; yet that advantage did not correspond to a statistically detectable improvement in the correctness or groundedness of the answers ultimately produced. There was significant influence by retrieval configuration on the quality of retrieval and the refusal rates,

but it did not correlate to a statistically noticeable difference in terms of correctness or groundedness of generated answers. This does not detract from the importance of improvements at retrieval level such as reranking; however, it should serve as a strong warning against using retrieval-level metrics as an adequate measure of generation-level quality. As RAG models continue to get more widespread use in knowledge-intensive applications, it becomes important to adopt better evaluation techniques that take into account this distinction and do not simply evaluate retrieval, but rather whether it translates into factual correctness of answers.

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